Pushing the Envelope
The Effects of Salary Negotiations

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Abstract

Salary negotiations are a widespread phenomenon that can shape labor market outcomes, including compensation and inequality. We provide novel empirical and theoretical insights into the causes and consequences of salary negotiations. We conducted a field experiment with over 2,000 U.S. job seekers in technology roles. We find that a light-touch encouragement treatment significantly increased both negotiation attempts and compensation gains, suggesting that information frictions cause some workers to leave money on the table. In the supplemental survey, we find that workers perceive significant risks from negotiating, including the possibility of having their job offers rescinded. We develop a new model of salary negotiations that incorporates information frictions, the risk of employer backlash, and strategic choices across employment options to better explain our experimental and non-experimental findings. Overall, our study suggests that reducing misperceptions and uncertainty around negotiation repercussions would meaningfully improve worker surplus.

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1 Introduction

Salary negotiations are a common feature of labor markets, with the potential to shape key outcomes such as employee compensation, pay inequality, employee-employer matches, and the gender wage gap. Yet despite their prevalence and potential importance, there is limited direct evidence on how salary negotiations unfold in practice and what consequences they have. In this paper, we help fill this gap by documenting novel facts about salary negotiations and developing a theoretical framework to interpret them.

When deciding whether to negotiate an offer, employees may expect, with some probability, to gain better compensation terms. Employees may also perceive risks from negotiating, a possibility that has received less attention. For example, employees may fear having the offer rescinded or the start date delayed. Indeed, when asked about the main barriers to negotiation, negotiation experts cited fear of retaliation as a key reason why individuals do not negotiate.[1] Thus, we consider a framework in which employees see negotiating as "pushing the envelope," or testing the limits of what is possible or acceptable.[2]

A survey of U.S. professionals by Fidelity (2022) provides a useful starting point: a majority (85%) of those who negotiate their job offers secure better compensation terms, often significant gains, but despite this, fewer than half (42%) attempt to negotiate. Given the success rate, why don't more employees attempt to negotiate? On the one hand, perhaps employees who do not attempt to negotiate are right not to—for example, if these are the employees who would be unlikely to secure gains if they asked. On the other hand, failing to negotiate could be a mistake—for example, some employees may overestimate the risks from negotiating. These information frictions could have important welfare and policy implications: employees who forgo negotiation because of such frictions may leave money on the table and end up with a worse employer match.

To study salary negotiations in a real-world setting, we designed an experiment for individuals actively navigating the job market. A baseline survey collected detailed information on subjects' backgrounds and allowed us to identify respondents who were actively searching for a job. A follow-up survey tracked key outcomes, including whether participants received

[1] More precisely, we conducted a small survey of negotiation coaches from levels.fyi. We posed an open-ended question asking why some individuals choose not to negotiate. Of the 11 coaches who responded, 6 explicitly mentioned fear of backlash.

[2] The expression refers to the concept of an envelope as a boundary, originating with test pilots who flew close to the boundary of an aircraft's *flight envelope*, which defines its operating limits.

a job offer, whether they attempted to negotiate it, and whether compensation improved from the initial to the final offer.

The baseline survey included an *encouragement treatment* designed to assess the role of uncertainty.[3] This treatment consisted of an additional screen with a narrative and information encouraging respondents to negotiate the terms of their job offers. This light-touch intervention included two statistics from the aforementioned survey (Fidelity, 2022): the share of individuals who attempt to negotiate and the share of those attempts that result in higher compensation. Prior to treatment assignment, we elicited subjects' beliefs about these two statistics. This enables us to examine heterogeneity according to whether subjects' prior beliefs were below or above the information provided in the experiment.

Under the hypothesis that some individuals do not negotiate because they overestimate the costs versus benefits of doing so, the treatment should make those employees more likely to negotiate. Moreover, the treatment should also improve their expected outcomes—otherwise, they would have been right not to negotiate in the first place. Our survey instruments included multiple questions designed to test this hypothesis. Most importantly, the follow-up survey allows us to measure the treatment effects on the probability of attempting to negotiate and on changes in compensation terms. Other survey questions paint a fuller picture of the costs and benefits. The follow-up survey included a broader range of outcomes, such as satisfaction with the negotiation and whether an offer was accepted. At the end of the baseline survey, we included a question about respondents' intentions to negotiate future offers, which provides a short-term outcome. The baseline survey also included questions about negotiation experience before treatment assignment, which can serve as placebo outcomes in an event-study fashion.

Our survey targeted workers in tech-related occupations across U.S. industries who were actively seeking jobs between May 2023 and April 2024. We recruited the sample in partnership with levels.fyi, a platform popular among workers with tech backgrounds seeking to compare positions across employers. We invited a large sample of its registered users to participate in our study if they were actively seeking job opportunities. A total of 2,399 subjects completed the baseline survey. Of these, 1,515 also completed the follow-up survey, and at that point 1,283 had received at least one job offer. In the control group, the average participant earned $215,700 annually and had 7 years of work experience. Participants worked

[3]The baseline survey included a second pre-registered arm—the *coaching treatment*—consisting of a deep discount on a professional negotiation coaching service. Due to space constraints, its results will be presented in a separate paper.

in industries such as information (37%), manufacturing (15%), finance and insurance (14%), and retail (10%). Common roles included Engineer, Product Manager, and Data Scientist, with top FAANG employers such as Amazon (7%) and Google (4%), as well as Walmart (1%) and Capital One (1%). Even though all respondents were active in the job market, most (71%) were already employed. Among those who received at least one job offer, 53% attempted to negotiate at least one offer, and 42% of those negotiators succeeded in increasing their compensation terms. Ultimately, 59% accepted one of the offers.

The distribution of prior beliefs provides suggestive evidence that information frictions are important. For example, beliefs about negotiation rates varied widely across subjects. Because participants assigned substantially different probabilities to the same underlying outcome, at least some of these beliefs must be inaccurate. Moreover, the errors do not appear to be symmetric: subjects were more likely to underestimate than overestimate both negotiation rates and the likelihood of successful negotiation.

The effects of the encouragement treatment also suggest that information frictions are a significant barrier to negotiations. The encouragement treatment significantly increased both the likelihood of negotiating and the probability of securing better compensation. More precisely, the treatment raised the rate of negotiation attempts by 7.9 percentage points (pp), from 53.3% to 61.2% (p=0.005), and the share who experienced compensation gains by 5.7 pp, from 30.0% to 35.7% (p=0.029).[4] Together, these results are consistent with the encouragement treatment inducing negotiations that generated compensation gains. They suggest that, consistent with information frictions, some subjects were leaving money on the table by not negotiating.

To provide more direct evidence about the role of information frictions, we split the sample according to whether subjects' prior beliefs about the prevalence and success of negotiations were below or above the information provided in the experiment. We refer to these two groups as the low-prior and high-prior groups, respectively. If the encouragement treatment operated by correcting biases in these prior beliefs, we would expect it to be more effective for the low-prior group (whose members find out that negotiations are more common and effective than they initially thought) than for the high-prior group. Consistent with this prediction, we find that the treatment effects are larger and statistically significant for the low-prior group, and close to zero and statistically insignificant for the high-prior group.

[4] As an alternative way to measure compensation gains, we show that the treatment raised the average offer improvement by 1 pp, from 4.02% to 5.02% (p=0.057), around a $1,700 increase relative to a control-group mean improvement of $10,000 (p=0.234).

One limitation of the analysis is that the response rate to the follow-up survey was 63% (and 53% completed it *and* received at least one offer), with a statistically significant difference between the treatment and control groups, which raises concerns about potential selection bias. We address this concern in multiple ways. For example, we show that the effects are significant for the intention to negotiate, which was measured at the end of our baseline survey and thus not subject to selection bias. We conduct a selective-attrition test proposed by Duflo et al. (2015), which takes advantage of the fact that we measured the intention to negotiate and past negotiation attempts at baseline. We also report BCGL bounds following Behaghel et al. (2015) alongside treatment effects for all follow-up survey outcomes. Taken together, the evidence suggests that selective attrition is unlikely to account for our findings.

One natural question is whether subjects in our main experiment were making mistakes or whether the risk of backlash justified their hesitation to negotiate. Backlash can take many forms, so to assess whether subjects experienced backlash, we examined a wide array of outcomes, including satisfaction with the decision to negotiate, likelihood of accepting an offer, and retention in the new job. Across all these metrics, subjects in the treatment group appeared to fare no worse than those in the control group. The point estimates are close to zero and statistically insignificant, and we can rule out negative effects greater than 10% across all endpoints. If anything, we find suggestive evidence of a positive rather than a negative effect: the encouragement treatment increased perceived alignment with the position. We are cautious, however, about interpreting these findings as definitive evidence that backlash did not increase.

To provide additional complementary evidence, we conducted a supplemental survey of 505 U.S. workers, also in partnership with levels.fyi and using a similar recruitment strategy. The survey revolved around a hypothetical job offer tailored to each respondent. We elicited beliefs about the potential costs and benefits of negotiating this offer. More precisely, we measured respondents' expectations about the effects of a negotiation attempt on a range of outcomes, including the probability of negotiation success, the probability of offer rescission, and the expected delay in starting the job. In one treatment arm, we randomized whether subjects considered requesting a 10% or 20% increase in compensation, to distinguish between uncertainty about *how much* to negotiate and uncertainty about whether they are *allowed* to negotiate in the first place. Additionally, we embedded the same encouragement treatment used in the main experiment to measure how this information affects the perceived costs and benefits of negotiating.

The results suggest that workers perceive meaningful costs to negotiating, including the risk of offer rescission, expected negotiation delay, and guilt. For example, respondents not shown the encouragement message expected that attempting to negotiate for 10% higher compensation would increase the probability that the offer would be rescinded by 6.46 pp relative to not negotiating. Moreover, the expected costs of negotiating increased with the size of the ask, but they did so at a markedly diminishing rate. Taken together, these findings support the view that some employees perceive that making counteroffers may not be allowed and thus that initiating a negotiation requires "pushing the envelope."

We also find that the encouragement treatment had significant effects on both the perceived costs and perceived benefits of negotiating, such as decreasing the expected negotiation delay and increasing the expected probability of success. Consistent with the main experiment, the effects on perceived costs were concentrated among subjects whose prior beliefs about the prevalence and success of negotiation were below the information provided by the encouragement message. This evidence reinforces the interpretation that the encouragement treatment induced additional negotiation attempts by mitigating information frictions.

Motivated by the evidence, we propose a theoretical model of salary negotiations, which we use as a lens to interpret our empirical results and explore the relevant factors that determine employees' negotiation behavior. Our model incorporates three key features that jointly account for the patterns we observe. First, employees are not passive price takers and can choose to counter employers' initial offers, even if doing so entails risks. Second, employees have incomplete information, and perhaps misspecified beliefs, about the cost versus benefit of negotiating—for example, they may overestimate the risk that a negotiation attempt leads to rescission of the offer. Third, employees potentially receive multiple job offers, framing bargaining under risk as a portfolio choice problem à la Chade and Smith (2006).

Upon receiving a job offer or several, an employee elects to negotiate by making counteroffers to any subset of them. She weighs the potential costs and benefits of negotiating with each employer conditional on her decision to bargain with others. For example, an employee may negotiate with a particular employer only if she has a sufficiently compelling "safety" offer from another employer with which she does not negotiate. We characterize the employee's optimal bargaining strategy and show that it depends on her beliefs about a given employer and the rest of her portfolio of options. Moreover, optimal bargaining follows a "top-down" structure: she initiates bargaining with the employers who have the highest match values

and negotiates with additional employers only if the expected gains exceed the inside option provided by the rest of her portfolio.

The model rationalizes our key experimental finding that the encouragement treatment increased the share of employees—particularly those with low prior beliefs—who attempt to negotiate: it changes their perceptions of risk. It also captures a series of non-experimental findings that differentiate our predictions from those of several canonical models. For example, our model allows for negotiations to fail (in our data, 60% of negotiation attempts result in no change in compensation terms). Because top match-value options are negotiated, our model predicts that bargaining can increase the dispersion of final compensation terms rather than decrease it, and that the final terms of an employee's two best offers will not match. In our data, negotiation increased dispersion in 54% of cases and decreased it in 41%, and only 19% of those with multiple offers end with their two top offers within $10,000 of one another.

Also, our model predicts and our data show that the strength of fallback options raises the probability of seeking negotiation. A 10 pp increase in the distance to the next-best option is associated with a 1.1 pp decline in the likelihood of negotiating. These patterns are not well described by an "offer matching" protocol,[5] nor by complete-information benchmarks that reduce bargaining to a single, immediately successful negotiation. Instead, they reflect interdependent portfolio choice: employees may preserve some offers as insurance while putting others at risk in pursuit of higher pay.

Our paper contributes to several strands of literature, offering both empirical and theoretical insights. On the empirical side, evidence on the role of salary negotiations remains limited. Card et al. (2016) use matched employer-employee administrative data to estimate a model that decomposes wage inequality into components attributable to sorting and bargaining. To get direct measures of negotiation rates, Hall and Krueger (2012) use survey data and find that roughly one-third of U.S. employees engage in wage bargaining.[6] More recently, Caldwell, Haegele, and Heining (2026) provide comprehensive survey evidence from Germany on the role of bargaining in labor markets. We contribute by uncovering novel evidence that information frictions and fear of employer backlash are key barriers to negotiation. We offer a bargaining microfoundation for firm-level wage premia, complementing Kline (2025) and

[5] Offer matching may be the predominant form that bargaining takes in other markets—most notably, universities are known for adjusting pay to match competing offers in academia.

[6] Salary negotiations are more common in high-skill positions (Hall and Krueger, 2012), presumably because these roles require more tailored offers (Cullen et al., 2026).

Card et al. (2018). Moreover, from a methodological standpoint, our study is related to Cortes et al. (2024) and Capozza (2024), who conduct experiments informing individuals about the gender gap in negotiations.[7]

On the theoretical side, our model contributes to the bargaining frameworks used to interpret wage outcomes. Standard approaches frequently assume that all relevant information is known (e.g., Nash, 1950; Rubinstein, 1982; Mortensen and Pissarides, 1994; Stole and Zwiebel, 1996; Postel-Vinay and Robin, 2002a,b; Cahuc et al., 2006). These complete-information frameworks suggest that there are no information barriers to negotiation and no failed negotiations in equilibrium, and they predict no effect of our experimental treatment. Our results show that several additional features not modeled in these frameworks—information, beliefs, and bargaining decisions with other employers—affect the decision to bargain with a particular employer and can influence where an employee ultimately works.

The importance of employee beliefs and the impact of our encouragement treatment complement the literature on bargaining with incomplete information (see, e.g., Myerson and Satterthwaite, 1983; Chatterjee and Samuelson, 1983; Fudenberg et al., 1985; Fuchs and Skrzypacz, 2010, and the literature surveyed in Ausubel et al. (2002)), which models information asymmetries as arising from not knowing the counterparty's match value. The reputational bargaining literature (e.g., Abreu and Gul, 2000; Atakan and Ekmekci, 2014; Fanning, 2016) instead models uncertainty about a counterparty's commitment to refraining from negotiations and, in particular, whether attempting to negotiate will cause backlash in the form of costly delay. Our theoretical framework and results are consistent with both of these modeling approaches, accommodating both uncertainty about offer negotiability and uncertainty about match values.[8]

The remainder of the paper is organized as follows. Section 2 describes the institutional context, experimental design, and implementation details. Section 3 presents the effects of the encouragement treatment. Section 4 presents evidence from our supplemental survey. Section 5 presents our model and discusses how it can account for key experimental and non-experimental findings that are difficult to reconcile with standard labor bargaining models. The final section concludes, and proofs of theoretical results are relegated to the appendix.

[7] For further discussion, see Supplement D.6, where we report the heterogeneity of the effects by gender.

[8] Closely related to our work, Friedenberg (2019) shows that strategic uncertainty can generate inefficiency in bargaining even when payoff fundamentals are known, through second-order optimism.

2 Experimental Design

2.1 Overview of the Experimental Design

We collected panel data, consisting of a baseline survey and a follow-up survey. The main goal of the baseline survey was to administer the treatment. The main goal of the follow-up survey was to measure the outcomes of interest, such as whether the subject attempted to negotiate. Sample baseline and follow-up survey instruments are attached as Supplements F and G, respectively. The survey was largely adaptive—for example, if a subject reported having an offer, we would ask additional questions about that offer, such as the compensation terms and whether he or she attempted to negotiate it.[9] The surveys typically took under 10 minutes to complete.[10] To encourage participation in the study, we paid subjects $50 via electronic gift card.[11]

The baseline survey started with a series of questions that were used as filters. To qualify for the study, subjects had to have permission to work in the U.S., be at least 20 years old, have a college degree, and either have an offer or be actively looking for one.[12] The survey collected identifiable information, including full name, email address, and employment history. In addition, we were able to match 84% of subjects to a LinkedIn profile.[13] The LinkedIn data allow us to validate survey responses, capture additional subject characteristics, and track subsequent job mobility—even for participants who did not complete the follow-up survey.

The baseline survey gathered detailed subject characteristics, including demographics, prior negotiation experience or training, and information about the respondent's current or most recent job—such as employer, job title, and compensation. To standardize compensation reporting, we closely followed the approach used by levels.fyi (see its salary submission form in Supplement J). Specifically, we asked separately about annual base salary, annual equity compensation, annual bonuses, and relocation or sign-on bonuses. Unless otherwise noted, *compensation* refers to total compensation: the sum of annual base salary (on average, 76.4% of the total), annual equity compensation (12.7%), annual bonus (6.1%), and other compensation (4.8%).[14] This definition implicitly assumes that employees value a dollar of each

[9] The full survey logic is detailed in Figures C.1 and C.2.

[10] Median completion time was 8 minutes for the baseline survey and 4.3 minutes for the follow-up survey.

[11] There were some small variations in these financial incentives—see Supplement C.1.

[12] For more details, see Supplement C.1. Additionally, we used data on IP addresses and LinkedIn profiles to confirm that the vast majority of subjects were indeed based in the U.S.

[13] We manually matched respondents to their LinkedIn profiles. See Supplement C.3.

[14] Other compensation includes sign-on and relocation bonuses. In practice, equity may vest unevenly over

component roughly equally—motivated by anecdotal evidence suggesting that employees primarily care about total compensation. Moreover, a simple analysis of subjects' choices between offers supports this approximation for the recurring components (base salary, equity, and bonus)—see Supplement C.6.

The goal of the follow-up survey was to track how job search unfolds—for example, whether subjects receive offers and attempt to negotiate them. One implementation challenge was the varying pace of job searches across subjects, so we staggered the invitation emails to accommodate this variation. Approximately six weeks after the baseline survey, we sent the first round of follow-up invitations. If a subject responded but had not yet received any offers, we sent a second invitation 45 days later. If they still had not received any offers by that second checkpoint, we sent a final invitation 45 days later. Subjects who did not respond to an invitation were sent multiple reminders. Among subjects who completed the follow-up survey, the median time between the baseline and follow-up surveys was 49 days.[15]

In the baseline survey, we asked subjects whether they had any job offers at the time. For those who did, we collected detailed information, including compensation terms. To keep the baseline survey manageable, we asked for details on only one offer—if they had multiple, we asked them to report their preferred one.[16] In the follow-up survey, participants were reminded about the offer they listed at baseline, and they were asked to list any additional offers and to provide details for each. We allowed subjects to list up to three new offers, including any offers from their current employer. A small minority of subjects (2%) had more than three additional offers; in those cases, we asked them to report their top three. For each offer reported (including the one from the baseline), the follow-up survey included a series of questions used to construct outcome variables, such as whether the subject attempted to negotiate and whether the compensation terms improved—see Section 2.3 below for more details.

2.2 Encouragement Treatment

The *encouragement treatment* was designed to primarily target what we call *negotiability uncertainty*: people may choose not to negotiate because they believe the employer is not

several years, so we asked subjects to report the average annual value across all vesting years.

[15] For more details, see Supplement C.2.

[16] Subjects with multiple offers at baseline were a minority: approximately 16.3% had two offers, and 9.2% had three or more.

open to negotiations.[17] This treatment consisted of a light-touch informational intervention that encouraged participants to negotiate job offers. Half of the subjects were randomly assigned to the encouragement treatment and saw an additional screen with the encouragement message (reproduced in Figure 1), while the other half—the control group—did not see this screen. In addition to a narrative encouraging negotiations, the message included factual information from a recent survey (Fidelity, 2022) stating that 42% of individuals attempt to negotiate offers and that 85% of those who do are successful.[18] The message also included additional arguments in favor of negotiating—for instance, that one should not feel guilty about negotiating and that employers expect it—as well as a mention of a Harvard Business School course highlighting the long-term financial benefits of negotiating.

Given that the treatment message included information about the share of individuals who negotiate and their success rate, it is natural—under standard learning models—to expect heterogeneous effects by prior beliefs. For example, individuals who underestimate the share of negotiators should update their beliefs upward in response to the information and thus become more likely to negotiate themselves. By contrast, individuals with accurate beliefs should not update and therefore should not respond to the information. Indeed, leveraging heterogeneity by prior beliefs is a common approach in information-provision experiments (see, e.g., Cullen and Perez-Truglia, 2022). With this in mind, we measured beliefs about the share of individuals who attempt to negotiate and their success rate immediately before administration of the encouragement treatment. These data allow us to explore the (pre-registered) heterogeneity by prior beliefs.

2.3 Outcomes of Interest

The first main outcome of interest is whether subjects attempted to negotiate. Specifically, the outcome variable equals one if the subject attempted to negotiate at least one offer and zero otherwise.[19] As with any survey question, our measure of negotiation attempts may suffer

[17]The experiment included a second arm, the *coaching treatment*, which offered a discount of 80%+ on a negotiation coaching service provided by levels.fyi. Assignment was cross-randomized: 25% received encouragement only, 25% coaching only, 25% both, and 25% neither. Due to space constraints, the design and results of the coaching treatment are presented in a separate paper. Although the two treatments could interact, low coaching take-up suggests that such interactions are unlikely to be quantitatively important: despite the deep discount, only 3.0% of subjects assigned to the coaching treatment signed up for the service.

[18]To enhance credibility, the treatment message included hyperlinks to all the referenced articles.

[19]Some subjects received multiple offers. For these subjects, the measure captures the extensive margin only—i.e., whether they attempted to negotiate any offer at all.

from some measurement error.[20] The second main outcome is whether the compensation terms improved, regardless of whether a negotiation attempt occurred. Specifically, the outcome variable equals one if the annual compensation increased from the original terms to the latest terms for at least one offer, and zero otherwise. We also consider two continuous versions of this outcome: the percentage improvement in compensation from initial to final offer, and the corresponding dollar change.[21]

There are two additional outcomes that are closely related to the negotiations outcome. First, we construct a measure that can pick up short-lived effects of the intervention. For instance, in the case of the encouragement treatment, subjects may have forgotten the information or it may have become less salient over time. At the end of the baseline survey—after subjects had been exposed to their assigned treatments—we asked whether they intended to negotiate in the future. The intention to negotiate outcome takes the value one if the subject expected to negotiate future offers almost surely, and zero otherwise.[22] Second, we included two questions that can serve as placebo outcomes for a falsification check. In the baseline survey, we asked subjects whether they had experience negotiating offers in the past. Because this question was asked before subjects were exposed to any treatments, we should expect no treatment effects on this outcome. Likewise, we asked another question—also administered before treatment assignment—on whether they had attempted to negotiate when they received the offer for their current or most recent job. This constitutes our second placebo outcome.

In addition to our main outcomes, the follow-up survey included a range of additional

[20]For example, some subjects may not pay attention to the question, may misread it, or may make mistakes when responding. In addition, there is subjectivity in how respondents interpret what constitutes a negotiation attempt. For instance, delaying a decision on an offer can be a negotiation strategy—the employer may grow anxious and improve the offer proactively. As a result, some respondents may report that they did not attempt to negotiate, even though they may have done so implicitly by merely waiting. Indeed, we observe some suggestive evidence consistent with this: among offers for which the subject did not report a negotiation attempt, a small but non-negligible share (13%) experienced an improvement in compensation terms.

[21]To limit the potential influence of outliers, we winsorize these outcomes at the 99th percentile. For subjects who experienced compensation improvements in multiple offers, we define the alternative outcome—to be consistent with the definitions of the other outcomes—as the maximum percentage change across all offers. Supplement D.1 shows that the results are similar when using the offer held at baseline instead. Compensation improvements are measured as the *net* change in total compensation, where total compensation is the sum of annual base salary, equity, bonuses, and other compensation. In the 13 cases where final compensation fell below the initial offer, we censor the percentage change at zero, reflecting our definition of this variable as a measure of improvement rather than overall change.

[22]We elicited responses to this question using a 1 to 4 scale ranging from "No" to "Almost Surely." For ease of interpretation, the analysis focuses on a binary indicator. However, results are similar when using the full cardinal scale.

measures intended to capture the broader impact of the interventions or underlying mechanisms. In particular, we measure in several ways whether encouraging negotiation carried costs for subjects. Two of these are reported alongside our main outcomes. The first is a survey measure of whether the respondent took the job: an indicator equal to one if the respondent accepted at least one offer. If negotiating led employers to withdraw offers or damaged the relationship with the employer, treated subjects should be less likely to take the job. The second is satisfaction with the negotiation, rated on a 1–5 scale (1 = "not at all satisfied," 5 = "extremely satisfied").[23] We complement the self-reported measure of taking the job with two measures built from LinkedIn profiles: whether the subject indicated switching jobs, including moves to a new role at the same employer, on LinkedIn[24] and whether they switched and remained at the new job for at least 12 months. The survey also measured several additional outcomes, which are described in detail in Supplement D.2.

2.4 Subject Recruitment

In this section, we describe the recruitment process for the main experiment's baseline survey, which began on May 25, 2023, and ended on October 10, 2023. We recruited the survey sample from individuals who had registered an account with levels.fyi, a website that is a common resource for job seekers as it allows them to compare the "levels" and position titles across firms. This let us target active job seekers in the United States—that is, individuals who had recently received a job offer or were currently interviewing and expecting to receive one in the near future. Individuals were required to register for a free account in order to access salary information on the levels.fyi website across different employers and positions. We sent email invitations to a sample of individuals who had recently registered an account.[25]

In the survey invitation (attached as Supplement K), we emphasized that the survey was intended for individuals actively participating in the job market. For those who did not respond,

[23] This outcome is, by construction, defined only for subjects who attempted to negotiate at least one offer. If subjects attempted multiple negotiations, we define the outcome as the highest satisfaction across all negotiation attempts.

[24] We count a move to a new role at the same employer as a switch, since tech workers commonly transfer across teams and departments. Because start dates entered in LinkedIn are not exact, we treat switches from 3 months before to 1 year after the baseline survey as moves associated with the job-seeking spell captured by our surveys.

[25] Each week we invited individuals who had signed up on the platform during that same week. Additionally, we would invite a random sample of individuals who had signed up during the previous year. From August 2, 2023, to August 15, 2023, we also sent invitations to individuals who expressed interest in the coaching service via a form embedded on levels.fyi's website. A small minority (0.43%) of subjects were recruited this way.

we sent reminder emails one week and two weeks later. Additionally, the registration form (Supplement I) included information that allowed us to target invitations to individuals who reported having U.S. work authorization and being active on the job market.[26]

A total of 2,399 subjects qualified for the study and completed the baseline survey.[27] Of those, 1,515 completed the follow-up survey, among whom 1,283 received at least one job offer.

2.5 Relevant Details About the Context

Our analysis focuses on compensation terms. This is of course not the only characteristic of the offer that matters to subjects, but it is arguably the most important. We find that a strong majority of subjects accepted the offer with the highest compensation.[28] For a more formal analysis, a companion paper (Cullen et al., 2025) uses a conditional logit model to show that compensation is the most important feature of the offer that predicts acceptance, although subjects also value other attributes such as remote work and the employer's reputation.

In our baseline survey, we asked whether the job offer was verbal or written; approximately 37% were verbal. This is consistent with anecdotal reports that employers often extend verbal offers first and only formalize them in writing once they believe the candidate is prepared to accept. This practice matters for two reasons. First, verbal offers may make negotiation attempts riskier, as employers can more easily withdraw them with fewer legal consequences.[29] Second, verbal offers make it more difficult to match outside offers, insofar as employers must rely on the candidate's word rather than written proof.

Another relevant feature of the context is the state of the job market during the study period. At that time, demand for software engineers was contracting. Hiring slowed significantly across employers, and there were widespread layoffs.[30] As a result, the returns to negotiations we

[26] More precisely, we selected individuals who, when asked about their job market status, chose one of the following options: "actively interviewing," "actively searching," or "offer in hand".

[27] We estimate that approximately 1.8% of those invited completed the survey (2.5% started it), somewhat below the 2.5%–4% reported by Giglio et al. (2021) among Vanguard clients receiving their first email invitation. The low completion rate is unsurprising given our survey's eligibility requirements—most importantly, that participants be active on the job market.

[28] For example, consider the 64 subjects who had no current job and exactly two outside offers that differed by at least 5% in compensation. In this group, 50 subjects accepted one of the two offers, and 82% of them accepted the offer with the higher total compensation.

[29] In the U.S., withdrawing a job offer is not automatically illegal because employment is usually at-will (Green, 2024). However, the employer could still be sued, for example, if the candidate claims that he or she was discriminated against.

[30] This likely reflected a combination of two factors. First, in the two years following the pandemic, tech em-

estimate in this study are possibly lower than they would have been in a stronger job market.[31]

2.6 Descriptive Statistics

Table 1 reports pre-treatment characteristics by treatment arm, both for the full baseline sample (columns (1)–(3)) and for the subsample that completed the follow-up survey and reported at least one offer (columns (4)–(6)), which is our main analysis sample. The average subject in the baseline survey control group was 31 years old and had 7 years of work experience. 71% of these subjects were employed at baseline. The average compensation (at their current or most recent job) was $215,700, compared with an average offer of $252,970 among control subjects at follow-up. Job offers thus tended to represent an opportunity to move to a better-paying position.

The sample consists of white-collar workers in tech-related occupations who are actively searching for jobs. The most common industries are information (37%), manufacturing (15%), finance and insurance (14%), and retail (10%) (see Table C.3).[32] Common job titles included Engineer, Product Manager, and Data Scientist, with top employers such as Amazon (7%), Google (4%), Walmart (1%), and Capital One (1%). About 19% of the sample identified as female, which is slightly below industry benchmarks—for example, Deloitte (2022) estimates that women hold approximately 25% of technical roles in the global tech workforce. About half of subjects had no prior experience negotiating salaries; while some had read a book or taken a course, most reported no formal training in negotiation.

Pre-treatment characteristics are balanced across arms. In the full sample, only one of ten differences is statistically significant (the share with a graduate or professional degree, 52.8% versus 48.7%, p=0.044), roughly what we would expect by chance, and none is significant in the analysis sample. The bottom panel of Table 1 reports attrition. Subjects in the encouragement group were more likely to respond to the follow-up survey (66.6% versus 59.7%, p<0.001) and to respond having received at least one offer (56.2% versus 50.8%, p=0.008).[33]

ployers may have hired too aggressively and subsequently had to scale back. Second, the rapid growth of artificial intelligence (AI) reduced the demand for software developers, particularly at the junior level (Hoover, 2025).

[31] Consistent with this, we surveyed levels.fyi negotiation coaches on whether offer negotiation had become harder in recent years. Of the 11 who responded, 10 reported that negotiating offers was more difficult in 2023–24 than in prior years, and 1 reported no change.

[32] See Supplement C.4 for a comparison of our sample with other benchmarks, such as the population from Hired.com described in Roussille and Scuderi (2025).

[33] Conditional on responding, the share reporting at least one offer is nearly identical across arms—84.4%(=100·56.2/66.6) in the encouragement group and 85.1%(=100·50.8/59.7) in the control group—so

These differences raise concerns about potential selection bias in the experimental estimates. For this reason, we present a range of robustness checks throughout the paper, including placebo outcomes and attrition bounds.

Turning to the follow-up survey, approximately 85.0% of control subjects had received at least one offer by the time of the follow-up survey.[34] Among them, 53% attempted to negotiate. Of those who negotiated, 42% experienced an improvement in compensation terms. Among the offers that improved, the average increase in total compensation was 13%, or $32,170. Ultimately, 59% accepted one of the offers.

3 Effects of the Encouragement Treatment

3.1 Distribution of Prior Beliefs

We begin by examining the distribution of prior beliefs, which is key to interpreting the effects of information. There was large heterogeneity in prior beliefs. For instance, some subjects believed that nearly everyone negotiates, while other subjects believed that nearly nobody negotiates. The fact that subjects gave widely different answers to a question with a single correct answer implies significant misperceptions. Whatever the true negotiation rate, it cannot match all of these differing beliefs—some subjects must overestimate it, others underestimate it, or both. For the interpretation of the effects of information, what matters most is whether the information provided to subjects was above or below their prior beliefs, which determines the direction of the belief updating and therefore the direction of the expected effects.[35] Relative to the statistics provided in the treatment, a majority of subjects (62%) underestimated the share of negotiators, while a vast majority (89%) underestimated the success rate. If the treatment provided information about a single statistic, splitting the sample would be straightforward. For example, if subjects were told only that the negotiation rate was 42%, we would distinguish between those with prior beliefs below 42%, who should revise their beliefs upward, and those with prior beliefs at or above 42%, who should leave their beliefs unchanged or revise them downward. Our setting is more complicated because the treatment provides information about

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the treatment difference arises at the response margin.

[34] 41.2% held exactly one offer and 43.8% held two or more (typically two).

[35] For a discussion of the accuracy of prior beliefs—including how the treatment statistics compare with our own survey results—see Supplement C.5.

two statistics—the negotiation rate and the success rate—so subjects may underestimate one while overestimating the other. Subjects who underestimated both statistics are assigned to the low-prior group, as they are expected to revise both beliefs upward. This group also includes subjects whose overall beliefs were low, defined as a combined index (the sum of their standardized beliefs about the two statistics) at or below the sample median. As a result, 70% are assigned to the low-prior group and the remaining 30% are assigned to the high-prior group.[36]

3.2 Average Treatment Effects

Table 2 summarizes the effects of the encouragement treatment. Each column corresponds to a different outcome and reports the average treatment effect, with the control-group mean at the bottom of the table. For now, we focus on the first row, which presents the results for the pooled sample, combining the low-prior and high-prior groups. The outcome variable in column (1) of Table 2 is the ex-ante intention to negotiate, which was elicited toward the end of the baseline survey. The encouragement treatment made subjects more enthusiastic about negotiating (at least in the short term), with an estimated positive effect of 6.3 pp (p=0.003) relative to the 45.2% of control-group respondents who reported that they would almost surely negotiate future offers.

The effect on the intention to negotiate should be interpreted with caution due to the hypothetical nature of the question and the potential for experimenter demand effects. While individuals may express a greater willingness to negotiate, they may not follow through when faced with real decisions.[37] For this reason, the key outcome of interest is whether, months later, subjects had actually attempted to negotiate when given the chance. The results for this outcome are presented in column (2) of Table 2. By construction, this outcome is defined only for subjects who responded to the follow-up survey and received at least one job offer—otherwise, they would not have had anything to negotiate.[38] In the control group, 53.3% of eligible subjects attempted to negotiate. The encouragement treatment increased this share by 7.9 pp (p=0.005).

Next, we examine the effects of the encouragement treatment on the probability of experi-

[36] On average, subjects in the low-prior group underestimate the negotiation rate by 15.6 pp (compared with an overestimation of 17.5 pp in the high-prior group) and they underestimate the success rate by 45.0 pp (compared with an underestimation of 19.0 pp in the high-prior group).

[37] Indeed, we observe that many subjects who stated at baseline that they would almost surely negotiate ultimately did not do so.

[38] Column (1) of Table 2 includes all subjects who completed the baseline survey and answered the intention question (N=2,321).

encing a compensation gain. Consider the marginal subjects who chose to negotiate as a result of the treatment. If their offer terms improved, this would suggest that not negotiating would have been a mistake. Conversely, if their negotiation attempts were largely unsuccessful, it would indicate that their initial reluctance to negotiate was justified.

Column (3) of Table 2 shows that the encouragement treatment increased the share of subjects with compensation improvements by 5.7 pp (p=0.029), from a control-group mean of 30%. This treatment effect on compensation improvements (5.7 pp) is nearly as large as the treatment effect on negotiation attempts (7.9 pp, from column (2)). Under assumptions that compensation improvements are mediated by negotiation,[39] the ratio of these effects implies that negotiating increased the probability of a compensation improvement by 72.8 pp ($=100\cdot\frac{0.05713}{0.07852}$) among subjects induced to negotiate by the treatment. Columns (4) and (5) of Table 2 show the size of the gains from negotiation, with the treatment increasing the percentage improvement in compensation by 1 pp (p=0.057) against a control-group mean of 4%. In dollar terms, the point estimate shows a \$1.7k improvement against a control-group mean of \$10k, though this is not precisely estimated (p=0.234).[40] Applying the same assumptions that compensation gains are mediated by negotiation (discussed in Supplement D.3), column (4) implies a 12.7 pp increase in the percentage improvement in compensation ($=100\cdot\frac{0.00999}{0.07852}$). This estimate suggests substantial gains from negotiating. To put this magnitude in perspective, we can compare it with estimates of wage markdowns in this labor market. Using initial offers, Roussille and Scuderi (2025) estimate a markdown of 19.5% in the U.S. tech sector, which is comparable to markdown estimates in a wide range of settings.[41] If an employee with an average markdown negotiated their offer, compensation would rise from 80.5% to 90.7% of their marginal revenue product of labor, or half the initial markdown.[42]

[39]This interpretation requires that encouragement did not discourage anyone from negotiating and that it affected compensation only through whether subjects negotiated. The latter assumption would fail if, for example, encouragement changed negotiation intensity among subjects who would have negotiated anyway and thereby affected compensation. Using the tests of Kwon and Roth (2026), we show our results are consistent with the compensation gains arising exclusively through negotiations. However, we cannot rule out changes in negotiation intensity. Supplement D.3 presents the tests in Table D.3 and Figure D.1. Supplement D.3 also presents the corresponding two-stage least squares (2SLS) estimates and discusses the role of these assumptions for their interpretation.

[40]As an alternative approach, we report unconditional quantile treatment effects in Supplement D.4.

[41]For example, Kroft et al. (2025) estimate a markdown of 20%, while Díez et al. (2025) find that the median markdown in the information sector is similar to the average across all sectors.

[42]A markdown of 19.5% implies that wages are 80.5% of the marginal revenue product of labor (MRPL). Holding MRPL constant, a 12.7% increase in compensation above the initial offer would raise this ratio to 90.7% ($=100\cdot0.805\cdot1.127$). The implied markdown would therefore fall from 19.5% to 9.3%, eliminating

Next, we examine whether encouraging negotiation carried costs for subjects. Although no single outcome captures all potential costs, we consider several measures that could reveal adverse consequences. Columns (6) and (7) of Table 2 report results for two such outcomes: offer acceptance and negotiation satisfaction. If negotiating provoked employer backlash, treated subjects might be less likely to accept an offer—because it was rescinded or the relationship with the employer was damaged—or at least they should report lower satisfaction with the negotiation. We find no effects on either outcome: the estimated effect on offer acceptance is close to zero (0.005, against a control mean of 0.591) and statistically insignificant (p=0.870); and the estimated effect on negotiation satisfaction is close to zero (0.038, against a control mean of 3.78) and statistically insignificant (p=0.665).[43] Neither estimate is statistically significant when we examine the low-prior and high-prior subsamples separately. We likewise find no evidence of adverse effects on additional outcomes, such as a LinkedIn-based measure of job mobility—see Supplement D.2 for details. If anything, the encouragement treatment increased self-reported alignment with the job (0.146 on a 1–5 scale, p=0.005).

3.3 Heterogeneity by Prior Beliefs

Table 2 also shows estimates of the treatment effect on the low-prior and high-prior groups. Because the factual information about negotiation rates and success rates was not the *only* content of the encouragement message, we cannot make unambiguous predictions about the direction of the effects for each group. However, the belief-updating mechanism yields a clear prediction about the relative magnitude of the effects: the treatment should have more positive effects for subjects in the low-prior group—who are updating both of their beliefs upward—than for those in the high-prior group, who are revising one or both beliefs downward.

Consistent with the belief-updating channel, Table 2 shows that the effects are more positive for the low-prior group than for the high-prior group. In fact, the effects were largely concentrated in the low-prior group. Column (1) shows that the treatment increased the intention to negotiate by 7.9 pp (p=0.001) for the low-prior group, compared to just 1.6 pp (p=0.666) for the high-prior group. Column (2) shows that the treatment effect on the share of negotiation attempts is 9.3 pp (p=0.005) for the low-prior group, but only 3.8 pp

approximately 52% of the initial gap.

[43] Satisfaction is observed only among subjects who negotiated. This comparison therefore provides suggestive evidence on whether subjects induced to negotiate had less satisfactory experiences than those who would have negotiated anyway.

(p=0.458) for the high-prior group. Column (3) shows that the treatment effect on the share with compensation gains is 7.4 pp (p=0.015) for the low-prior group, compared to almost no effect for the high-prior group (0.1 pp, p=0.984). Similarly, column (4) shows nearly no effect for the high-prior group (p=0.998) but a 1.3 pp increase in percentage improvement in offers in the low-prior group (p=0.025). In dollar terms, this is approximately a $2.7k increase (p=0.056), in contrast to an imprecise estimate that is statistically indistinguishable from zero for the high-prior group (p=0.570). Because effects for the high-prior group are imprecisely estimated, the differences between low- and high-prior groups are individually statistically insignificant. Nonetheless, when taken jointly, the consistent direction and magnitude of the effects across all outcomes are reassuring.[44]

The control-group outcomes help put the differences in treatment effects between low- and high-prior subjects in perspective. As columns (2)–(5) of Table 2 show, subjects with low priors in the control group were less likely to negotiate and to experience compensation gains than those with high priors. By closing the gap in beliefs, the encouragement treatment mostly closed the gap in negotiation attempts and closed about half of the gap in the share with compensation gains. In sum, the large heterogeneity by prior beliefs suggests that the factual content of the encouragement treatment played a central role in driving the observed effects. It also provides reassurance that the treatment effects are genuine rather than spurious.

We next examine whether encouraging negotiation can mitigate existing labor-market disparities. In particular, lower-earning workers may earn less in part because they have been less likely to negotiate, while gender differences in negotiation behavior are often cited as a contributor to the gender pay gap (e.g., Sandberg and Scovell, 2013). We therefore study treatment-effect heterogeneity by baseline earnings and gender, with a focus on whether the treatment narrows pre-existing gaps along these dimensions. The results are presented in Supplement D.6 and summarized below. The effects of the encouragement treatment are stronger for low-earners than for high-earners. The treatment narrows the gap in negotiation rates between these two groups by approximately 13 pp (p=0.017), closing about 80% of the pre-existing gap. These effects are accompanied by corresponding improvements across our other endpoints, including compensation, job switching, and alignment. We also find suggestive evidence that the encouragement treatment was more effective for women than for men, consistent with the possibility that encouraging negotiation can help narrow gender differences in negotiation behavior.

[44]Within the low-prior group, all four main follow-up effects remain statistically significant after adjusting for multiple hypothesis testing—see Supplement D.5 for details.

3.4 Additional Robustness Checks

One concern is the potential for selection bias, as some outcomes are measured for a subsample of subjects—specifically, those who responded to the follow-up survey and had at least one offer. If the treatment affected either the response rate or the composition of respondents, this could bias the results. While we observe some differences in response rates across treatment groups (see Table 1), we find no evidence of composition effects: treatment and control groups remain balanced on pre-treatment characteristics even within the restricted sample of subjects with at least one offer at follow-up.

We address concerns about selection bias in several ways. First, we report BCGL bounds (Behaghel et al., 2015) alongside treatment effects for all follow-up survey outcomes. This method leverages information on the number of contact attempts and is therefore tighter than standard trimming bounds. For the low-prior group, the encouragement treatment effect on all four main follow-up outcomes (columns (2)–(5) of Table 2) remains bounded away from zero.[45]

Second, we use the intention to negotiate future offers, which was measured at the end of the baseline survey for the full sample and is therefore not subject to attrition. The treatment effects on this ex-ante intention are qualitatively similar to those on ex-post negotiation behavior. Moreover, following Duflo et al. (2015), we estimate the treatment effect on this outcome separately in the full baseline sample and in the follow-up subsample. The two estimates are similar in magnitude and statistically indistinguishable—see Supplement D.8 for details—providing reassurance that our findings are not driven by selective attrition.

Third, we present a series of falsification tests using pre-treatment negotiation outcomes, in the style of an event-study analysis. We estimate these pre-treatment "effects" in both the full sample and the follow-up subsample (i.e., those who had at least one offer at follow-up). If the post-treatment effects are driven by selection bias, we would expect to observe similar effects in the pre-treatment outcomes. The results, presented in Supplement D.8, suggest that our findings are not driven by selection bias.

Fourth, the pattern of differential attrition across prior-belief groups provides additional reassurance. The treatment differences in follow-up participation are, if anything, larger for high-prior than for low-prior subjects (11.3 pp versus 4.9 pp; p=0.135 for the difference)—see Supplement D.9 for details. Yet the estimated treatment effects on negotiation and compensation gains are concentrated among low-prior subjects. This contrasting pattern provides

[45] See Supplement D.7 for a comparison of the BCGL bounds with the Lee bounds (Lee, 2009).

additional evidence against selective attrition as the sole explanation for our findings. Fifth, controlling for pre-treatment characteristics, which can mitigate selection bias on observables, yields similar results—see Supplement D.10 for details.

4 Supplemental Survey

We fielded a supplemental survey of 505 U.S. workers, again in partnership with levels.fyi and using a similar recruitment strategy. A sample of the full survey instrument is reproduced in Supplement H and summarized below.[46] The survey serves two purposes. First, it examines information frictions by measuring how the encouragement treatment changes beliefs about the costs and benefits of negotiating. Second, it provides direct descriptive evidence on workers' perceptions of the risks associated with negotiation.

4.1 Survey Design

The Hypothetical Offer. One key difference between the main experiment and the supplemental survey is that the latter revolved around a hypothetical job offer. Although hypothetical, the offer was tailored to each respondent to make it realistic. We asked respondents to name a specific employer and a position there for which they were qualified and which they would seriously consider accepting. We then asked them to report the lowest and highest plausible annual compensation they could expect for that role at that employer. Total compensation includes salary, equity, and bonuses. We set the hypothetical offer at 25% of the distance between the lower and upper bounds of the respondent's stated compensation range, leaving room for the respondent to negotiate the offer upward—see Supplement E.2 for details.

Encouragement Treatment Arm. The first treatment arm replicated exactly the encouragement treatment arm from the main experiment. Half of subjects saw the same information screen, reproduced in Figure 1. The main goal was to measure whether and how the encouragement treatment affects the perceived costs and benefits from attempting to negotiate. As in the main experiment, we measured prior beliefs about the share of job seekers who negotiate and the share of negotiators who succeed.

[46] Additionally, Supplement E.1 provides an overview of the survey logic.

Ask-Size Treatment Arm. The survey included a series of questions about the perceived costs and benefits of negotiating the offer. The second treatment arm randomized the size of the compensation increase respondents were asked to consider: either 10% or 20%.[47] These two values were calibrated using the distribution of gains observed in the main experiment so as to represent a typical versus a more aggressive counteroffer. More precisely, among respondents who negotiated and obtained an improvement, a 10% ask is close to the median gain (10.4%), while a 20% ask is exactly the 80th percentile of the gain distribution. We made the assigned ask size explicit in each question, using the language "attempting to negotiate for a [10%/20%] increase."

Outcomes of Interest. We elicited a series of perceived costs and benefits of negotiating the offer. The first main outcome concerns offer rescission. We elicited the perceived increase in the probability that the offer would be rescinded if the respondent attempted to negotiate, relative to making no attempt.[48] Likewise, we elicited the perceived effects of a negotiation attempt on the delay in starting the new job (in weeks), the employer's assessment of the respondent's fit, the guilt the respondent would feel, and overall negative consequences. Finally, respondents were asked about the probability that the negotiation attempt would be successful.

Recruitment and Sample. We recruited individuals who had recently registered an account with levels.fyi, resided in the United States, and had not completed the survey for the main experiment. Participation was compensated with a $20 electronic gift card, subject to professional-profile verification and response-quality checks. Our analysis sample contains 505 verified respondents, screened using checks for automated or inattentive responses and manual validation of professional profiles. The supplemental survey sample is broadly similar to the main experimental sample on observable characteristics.[49] Supplement E.5 provides additional details on recruitment and data-quality checks and shows that pre-treatment characteristics are balanced across treatment groups.

[47] The two treatment arms were cross-randomized, yielding four groups, each comprising ~25% of subjects—for example, 10%-ask with encouragement treatment, 20%-ask without encouragement treatment, and so on. See Supplement E.3 for details on treatment assignment.

[48] To make the elicitation cognitively accessible and clear, we elicited this belief in three stages—see Supplement E.4 for details.

[49] See Supplement C.4 for the comparison. One significant difference is a higher employment rate in the supplemental survey sample, likely reflecting differences in eligibility criteria: the main experiment required active job search, whereas the supplemental survey did not.

4.2 Results

The first result is that workers perceive meaningful costs to negotiating. These descriptive results are presented in Figure 2. Each panel corresponds to a different question. This figure reports outcomes for our reference group: subjects assigned to the 10%-ask and *not* assigned to the encouragement treatment. Panel A corresponds to the perceived risk of offer rescission.

Even if no negotiation took place, the average respondent thought that there was an 8.7% probability the offer would be rescinded.[50] Most importantly, a significant share (46%) of respondents expected that attempting to negotiate would increase the probability that the offer would be rescinded. On average, respondents expected that a negotiation attempt would increase the rescission probability by 6.46 pp. This constitutes a 74% increase relative to the perceived probability without negotiation, from 8.7% without a negotiation to 15.1% with a negotiation attempt. This average masks meaningful heterogeneity, with some individuals expecting a small increase while others expected a sharp increase.

Our survey also included a broader open-ended question asking respondents to describe the worries that made them hesitant to negotiate a past job offer. These responses indicate that rescission risk is the most common concern, mentioned by 39% of respondents.[51] Offer rescission does occur in practice: for example, in a U.S. survey, 16% of job seekers reported having had an offer rescinded (Bachaud, 2026). As discussed in Section 2.5, offers can often be withdrawn with limited legal consequences, particularly when they are verbal. At the same time, workers may overestimate the risk that negotiating will jeopardize an offer. Consistent with this possibility, Hart et al. (2024) find that, in surveys of the academic job market, job candidates perceived a substantially higher likelihood that negotiation would jeopardize an offer than respondents with experience on hiring committees.

Respondents also perceived other risks from negotiating an offer besides rescission. Panel B shows that, on average, subjects expected that a negotiation attempt would delay the job start by 1.35 weeks. Panel C shows that the majority (60.5%) of respondents would feel a little guilty or moderately guilty about attempting to negotiate the offer.[52] Panel D presents the

[50] For more details, see Supplement E.4; Supplement E.7 shows the full distributions of these beliefs.

[51] We used a large language model (LLM) to classify these responses. Some of the other common categories are a concern about making a negative impression on the employer (21% of respondents), feelings of self-doubt (19%), and hiring delays (5%)—see Supplements E.8 and E.9 for more details about the methodology and results.

[52] Relatedly, Gago (2026) provides evidence from a laboratory experiment that the emotional costs of confrontation can deter individuals from engaging in otherwise profitable negotiations, particularly when bargaining takes place in person.

results from a question on a negotiation attempt may affect the employer's assessment of the employee's fit with the position. Interpreting this outcome is tricky because negotiating can be perceived either positively or negatively. A majority (51.0%) did not think that attempting to negotiate would affect the employer's assessment of fit, with 19.1% thinking it could worsen this assessment and 30% thinking it could improve it. Panel E corresponds to a more general question about concerns with backlash, on a 5-point scale from "not at all" to "extremely." Only a minority (31.3%) of workers reported not being concerned at all. This is broadly consistent with the responses to the open-ended question, according to which 17.8% expressed no concerns about negotiating.

While the other panels capture potential risks from negotiation, the last panel is about the potential benefits. Panel F shows that, on average, the perceived probability that the 10%-ask is successful is 39.7%. This average masks significant heterogeneity, with some individuals thinking the probability is close to 0% and others thinking it is closer to 90%.

Table 3 reports the results from the two treatment arms.[53] Panel A compares respondents randomly assigned to a 10% ask with those assigned to a 20% ask, pooling across the encouragement arms. The first two rows report the mean of each outcome in each ask group, and the third row reports the difference (20% minus 10%). In column (1), the dependent variable measures offer rescission risk. Respondents assigned to the 10% ask expected negotiating to raise the rescission probability by 5.74 pp on average, and those assigned to the 20% ask by 8.73 pp. Doubling the ask therefore raised the perceived rescission risk by a further 2.99 pp (p=0.008). In other words, about two-thirds of the perceived rescission risk of a 20% ask is already present at a 10% ask. Relative to the 10%-ask, the 20%-ask increased guilt (column (3); 0.179, p=0.050) and overall concern (column (5); 0.240, p=0.006). For these two outcomes, the marginal cost of moving from a 10% to a 20% ask is small compared with the marginal cost of moving from no negotiation to a 10% ask.[54] In column (6), the dependent variable is the perceived probability that the employer accepts the ask. Naturally, the probability of acceptance declined with the size of the ask, but the magnitude of the decline was relatively modest: from 41.3% to 35.1%, a difference of 6.2 pp (p<0.001).

Next, we discuss the effects of the encouragement treatment, which are presented in Panel B of Table 3. As in the main experiment, we present the results for the whole sample (first row) as

[53] These results are robust to adding control variables; see Supplement E.6.

[54] For instance, the overall concern was 1.02 points (on the 0–4 scale) with the 10% ask, and it went up by only 0.240 points (or about 24%) after doubling the size of the ask from 10% to 20%.

well as separately for the *low-prior* and *high-prior* groups (second and third rows).[55] Overall, we find evidence that the encouragement treatment shifted many of these beliefs. Consistent with the main experiment, these effects were concentrated among respondents whose prior beliefs about the prevalence and success of negotiation were below the information provided by the encouragement message, in that we generally find large and statistically significant effects in the low-prior group but effects that are closer to zero and statistically insignificant for the high-prior group.[56]

Column (1) of Table 3 shows a negative effect of the encouragement treatment on the perceived rescission risk. On the one hand, the effect is large in magnitude: e.g., in the low-prior group, the encouragement treatment decreased the perceived rescission risk by 2 pp, which amounts to 33% of the reference-group mean of 6.14 pp for this group. On the other hand, this effect is imprecisely estimated (p=0.130). Column (2) shows that the encouragement treatment had a negative effect on expected negotiation delay. For example, in the low-prior group, the treatment reduced the expected delay by 0.276 weeks (p=0.039), which amounts to about 19% of the reference-group mean of 1.43 weeks. Column (3) shows that the encouragement treatment had a positive and significant effect on expected guilt from negotiating. The encouragement message reads "Don't feel guilty about negotiating", so we were expecting a negative rather than a positive effect on this outcome. Our preferred interpretation is that this is a reporting rather than a real effect on guilt: by normalizing the fact that people feel guilty, the message makes respondents feel more comfortable revealing that they feel guilty. Column (4) shows that the encouragement treatment had a negative and statistically significant effect—that is, the encouragement treatment shifted the perceived effect of negotiating on the employer's assessment of fit in a more favorable direction. The effect is statistically significant but more modest in magnitude: e.g., in the low-prior group, the effect of -0.286 points (on the 0–4 scale) amounts to 16% of the reference-group mean of 1.84 points. Column (5) shows no significant effects on the subjective measure of overall concern about backlash.

[55] We split the sample into low-prior and high-prior groups using the same criteria as in the main experiment. In this supplemental survey, 72.3% of subjects are assigned to the low-prior group, and the remaining 27.7% to the high-prior group. On average, subjects in the low-prior group underestimate the negotiation rate by 17.3 pp (compared with an overestimation of 17.4 pp in the high-prior group) and they underestimate the success rate by 46.1 pp (compared with an underestimation of 21.5 pp in the high-prior group).

[56] As in the main experiment, we note that since the effects for the high-prior group are imprecisely estimated, the differences between low- and high-prior groups as estimated by an interaction regression are largely individually statistically insignificant. The direction of the heterogeneity matches the main experiment for the rescission, delay, and fit outcomes, but not for guilt and overall concern.

Turning to perceived benefits, column (6) of Table 3 shows that the encouragement treatment had a positive impact on the expected benefits from negotiating, via an increase in the perceived probability of a successful negotiation. There are two caveats with this finding, however. First, the magnitude of the effect is more modest: e.g., in the low-prior group, the treatment raised the perceived success probability by 3.8 pp, which amounts to 10% of the reference-group mean of 37.5%. Second, unlike the effects in the main experiment, we find no heterogeneity between the low-prior and high-prior groups (e.g., point estimates of 3.807 and 3.392, respectively).

In sum, the results from the supplemental survey help explain why the encouragement treatment induced additional negotiation attempts in the main experiment. That is, the effects of the encouragement treatment in the main experiment are likely due to a reduction in the perceived costs of negotiating and—although less clear—an increase in the perceived benefits.

5 The Model

5.1 Outline of the Model

Our field experiment on salary negotiations reveals three key building blocks for our theoretical framework. First, employees have bargaining power in that they actively choose to make counteroffers to employers. Second, there exists incomplete information: in our base model presented below, an employee does not know if an employer's initial job offer is "negotiable." Attempting to negotiate a non-negotiable offer results in negative consequences. Third, employees potentially have multiple job offers, making their fallback option endogenous to negotiation decisions with other employers.

These features have important implications for efficiency in the labor market, because asymmetric information can lead to inefficient outcomes. Similar to Akerlof's (1970) lemons model, information frictions can prevent matches between an employee and an employer with a high value of labor from forming—employees may fail to negotiate when they should have, ending up at an employer with a lower value of labor. On the other hand, employees may push for higher compensation when they shouldn't have and lose a match entirely. An important distinction from classic models is that with multiple offers, employees face a portfolio optimization problem (as in Chade and Smith, 2006) because they each seek only one job. This implies that the decision to negotiate with one employer depends both on the other offers the

employee receives and on her choices to negotiate these other offers.

We characterize an employee's optimal bargaining strategy as a function of her portfolio of options and her beliefs. Our model, designed to capture key information frictions from our experiment, yields striking predictions for negotiation patterns that distinguish it from other labor bargaining models. We describe broad support for our model in the observational data.

Before presenting our bargaining framework, we note that our base model allows only for negotiability uncertainty—attempting to negotiate an offer may yield backlash, regardless of the amount demanded—but not uncertainty about the match value with each employer. This choice is made for simplicity of exposition and based on our findings presented in Section 4.2, which show that the risk associated with a relatively modest counteroffer is large compared to the marginal risk of additional wage demands. Negotiability uncertainty implies a relatively large perceived chance of backlash even at modest requested increases, while match-value uncertainty implies a smoother increase in risk as the size of the counteroffer increases. Our base model therefore considers this empirically relevant negotiability uncertainty friction, and in Appendix A.3, we extend the model to allow for match-value uncertainty as well, and we show that our results hold in that extension.

5.2 Setup

There is a single employee with an outside option denoted $\varnothing$ with value (or "wage") $v_\varnothing$ normalized to zero. There are $N \geqslant 1$ employers forming set $\mathcal{N}:=\{1,2,...,N\}$. Each employer makes the employee a job offer, one of which may be the employee's current employer. The employee generates a commonly known match surplus $v_i \in \mathbb{R}$ if matched with employer $i \in \mathcal{N}$,[57] where we assume that $v_i \neq v_j$ for all distinct employers $i,j \in \mathcal{N}$. Each employer i offers a wage $w_i \leqslant v_i$, where $w_\varnothing = 0$.

Upon receiving a vector of initial wage offers $\{w_i\}_{i\in\mathcal{N}}$, the employee can elect to engage in simultaneous negotiations with employers by making counteroffers. Formally, the employee simultaneously selects an additional wage demand $o_i \geqslant 0$ for each employer $i \in \mathcal{N}$; if $o_i > 0$ then we say that the employee *makes a counteroffer* to (or *negotiates* with) employer i. For accounting purposes, we fix $o_\varnothing = 0$.

[57] We do not take a stance on the genesis of each match value v_i, and treat it as exogenous. It represents the entire surplus generated from the worker's employment at employer i if the employer has no prospect of filling the position with another worker, or if employer i has an outside option of instead hiring a part-time worker at a known rate, v_i would represent the surplus over and above this outside option.

Making a counteroffer to employer i potentially exposes the employee to negative consequences for two reasons. First, i will withdraw the offer if the employee demands more than the match value, $o_i > v_i - w_i$. Second, i's offer may be *non-negotiable*, in which case making a counteroffer results in i's job offer being withdrawn regardless of its amount $o_i > 0$. The employee has *negotiability* uncertainty, captured by the vector $\hat{\rho} = (\hat{\rho}_1, ..., \hat{\rho}_N)$, where $\hat{\rho}_i$ specifies the employee's belief that employer i's offer is negotiable. Negotiability is independent across employers. Therefore, we denote an employee's *portfolio of offers* from employers in $\mathcal{N}$ as $\{w_i, v_i, \hat{\rho}_i\}_{i \in \mathcal{N}}$, noting that a "portfolio" contains a single offer when $N = 1$.

Given a portfolio and a vector of counteroffers o, let $\tilde{w}_i$ be the final wage available from matching with $i \in \mathcal{N} \cup \{\varnothing\}$:

$$\tilde{w}_i(o) = \begin{cases} w_i & \text{if } o_i = 0, \\ -\infty & \text{if } o_i > v_i - w_i, \text{ and} \\ \begin{cases} w_i + o_i & \text{with probability } \hat{\rho}_i, \\ -\infty & \text{with probability } (1 - \hat{\rho}_i) \end{cases} & \text{if } o_i \in (0, v_i - w_i] \end{cases}$$

Thus, $\tilde{w}_\varnothing = 0$ and the employee is hired by an employer $i^* \in \underset{i \in \mathcal{N} \cup \{\varnothing\}}{\operatorname{argmax}} \tilde{w}_i$ and receives a final payoff equal to her wage $\tilde{w}_{i^*}$.

This assignment procedure has the following interpretation. The employee accepts a job at an employer paying the highest wage. The wage employer i pays is equal to the initial wage offer ($\tilde{w}_i = w_i$) if $o_i = 0$. If $o_i > v_i - w_i$, then this offer drops in value below that of the employee's outside option ($\tilde{w}_i = -\infty$) because the offer is rescinded.[58] Otherwise, the employer accepts the terms of the employee's counteroffer ($\tilde{w}_i = w_i + o_i$) if the original offer was negotiable and rescinds the offer ($\tilde{w}_i = -\infty$) if the original offer was non-negotiable.

Robustness and Discussion of Assumptions. Our model makes a number of simplifying assumptions. Here, we discuss four extensions to our model that do not qualitatively change our upcoming takeaways.

First, our assumption that a failed negotiation leads to offer rescission can be relaxed. Specifically, even if the job offer remains available, our empirical evidence suggests that employees are concerned that some of the match surplus may be destroyed due to a deterio-

[58] Therefore, the normalization that $\tilde{w}_i = -\infty$ if the offer is rejected is merely a mathematically convenient way to say the employee will never be assigned to an employer that rejects her counteroffer.

ration of the interpersonal relationship, or a delay in onboarding. Additionally, the "fit" with a job can be affected if, for example, different job responsibilities are assigned conditional on negotiating. A reduced-form representation to broadly capture "intermediate" predictions and under which our model naturally extends is as follows: rejected counteroffers result in the original offer being available but its value eroded by a (potentially random) amount, and "accepted" counteroffers are available at the updated terms minus a (potentially differently distributed) erosion factor. Our base model corresponds to full erosion of value following a rejection, and no erosion following a success. However, our central intuitions carry over to the case that making a counteroffer exposes an employee to a distribution of surplus loss from the job offer, which potentially depends on the outcome of the negotiation.

Second, our assumption of simultaneous negotiations can be relaxed. More realistically, job offers are negotiated sequentially, and negotiating plausibly commits the employee to accepting that job, i.e., the employee may need to declare, "if you accept my counteroffer, I'll sign on the dotted line right now," in order for the employer to take the counteroffer seriously. The solution to our simultaneous-negotiation game is equivalent to that of a sequential-negotiation game in which the employee selects an order in which to negotiate with the employers, and she is matched to the first employer that either accepts her counteroffer, or to whom she makes no counteroffer (i.e., she accepts the employer's initial offer).[59]

Third, our assumption that employees have full bargaining power at the counteroffer stage—they make take-it-or-leave-it offers to employers—can be relaxed so that counteroffers result in split surplus conditional on a match. For a given portfolio of offers and an employer i, let $\lambda_i \in [0,1]$ be the Nash bargaining weight with employer i. We do not microfound this weight, as, following Binmore et al. (1989), it may be a function of (the employer's knowledge of) other offers available to the employee. As in our current model, if the employee elects to bargain with employer i, she believes there is a $1-\hat{\rho}_i$ chance that the job offer will be rescinded. If the offer is not rescinded, the final wage term denoted $\hat{v}_i$ splits the difference between the initial offer and the match value: $\hat{v}_i := (1-\lambda_i)w_i + \lambda_i v_i$. Our base model is the case in which $\lambda_i = 1$ for every portfolio and every employer i. Our analysis is otherwise unchanged under this alternative, appropriately replacing v_i terms with $\hat{v}_i$ terms.

Finally, our findings extend if the employee has uncertainty about her match value with

[59] We highlight that the dynamics we study involve the order of negotiations, not the effects of dynamism within a particular bilateral negotiation, as, for example, is studied extensively in the "Coase conjecture" literature (Gul et al., 1986; Gul and Sonnenschein, 1988; Ausubel and Deneckere, 1989, 1992a,b).

employers.[60] Analysis of such a model requires sufficiently different machinery, which we explicitly study in Appendix A.3, with proofs in Supplement B. Therefore, our qualitative predictions hold in markets where employees are unsure of whether negotiating will lead to backlash, are unsure of how much they can ask for without triggering backlash, or are unsure of both.

5.3 Optimal Portfolio Bargaining

For a given portfolio of wage offers $\{w_i, v_i, \hat{\rho}_i\}_{i \in \mathcal{N}}$, we seek an optimal vector of counteroffers to each employer $o^* := \{o_i^*\}_{i \in \mathcal{N}}$ that satisfies:

1. o^* maximizes the employee's expected (given $\hat{\rho}$) payoff,[61] and

2. Only "serious" counteroffers are made; if the employee has a zero probability of matching with an employer $i \in \mathcal{N}$ under any tie-breaking rule according to criterion 1, then $o_i^* = 0$. Formally, employer i is *payoff relevant* under a vector of counteroffers o if it belongs to the set of payoff-maximizing available options with strictly positive probability.

$$\mathcal{R}(o) := \left\{ i \in \mathcal{N} : \Pr\left(i \in \underset{j \in \mathcal{N} \cup \{\varnothing\}}{\operatorname{argmax}} \widetilde{w}_j(o) \right) > 0 \right\}$$

   denotes the set of payoff-relevant employers under o. Only serious counteroffers at an optimal o^* require $o_i^* = 0$ if $i \in \mathcal{N} \setminus \mathcal{R}(o^*)$.

The only "serious" counteroffers condition cuts down on a multiplicity of counteroffer vectors that do not affect the final outcome: workers do not negotiate with an employer if doing so will surely not result in matching with that employer. Intuitively, if the employee experienced a fixed marginal cost of negotiating with each employer, this condition would necessarily be satisfied. Our solution concept mimics the limit as such marginal negotiation costs go to zero.

An initial observation is that if an employee negotiates with an employer $i \in \mathcal{N}$ then the optimal counteroffer equals $v_i - w_i$; demanding the entire match surplus maximizes the wage available at employer i conditional on the offer being negotiable, because any lower counteroffer is accepted by the employer, while any higher counteroffer ensures rejection.

[60] Without match-value uncertainty, our upcoming analysis finds that the employee is more likely to negotiate with a particular employer if her "inside option" is larger. With unknown match values, this is strengthened: both the likelihood of negotiating and the amount of her counteroffer are increasing in her "inside option."

[61] Allowing for risk-averse or risk-loving preferences does not change our qualitative results.

Remark 1. *The optimal counteroffer for each employer* i *satisfies* $o_i^* \in \{0, v_i - w_i\}$.

The optimal counteroffer portfolio trades off risk—captured by $1-\hat{\rho}_i$ values—and reward—captured by $v_i - w_i$ values. Intuitively, the smaller $\hat{\rho}_i$ is and the smaller $v_i - w_i$ is, the less likely the employee is to make a counteroffer to employer i (i.e., she is more likely to set $o_i^* = 0$). For extreme values, this decision is simple: if $\hat{\rho}_i = 0$ then the employee is not willing to make a counteroffer to employer i regardless of the value $v_i - w_i$, because the employee perceives no chance of success. On the other hand, if $\hat{\rho}_i = 1$ and $v_i > w_i$, demanding $v_i - w_i$ involves no risk of rejection and weakly increases payoff (of course, the serious-counteroffers condition still rules out counteroffers to payoff-irrelevant employers.)

For values of $\hat{\rho}_i \in (0,1)$ an additional complication arises when constructing the optimal counteroffer to employer i: the risk versus reward calculation is affected by the presence of other offers and the endogenous negotiation decisions the employee makes for these offers. Intuitively, even given a relatively low value of both $\hat{\rho}_i$ and $v_i - w_i$, the employee will be willing to negotiate with employer i if she has a solid backup offer. For example, consider a portfolio with only two offers i and j, where $v_i > w_i > 0$, $\hat{\rho}_j = 0$, and $w_j = w_i - \epsilon$ for sufficiently small $\epsilon > 0$. In this case, the employee will optimally not negotiate with employer j because $\hat{\rho}_j = 0$, so the presence of j's initial offer serves as a risk-free alternative. The employee will optimally "roll the dice" and negotiate with employer i since her effective outside option w_j is only slightly worse than what she gets from not negotiating with i. On the other hand, for any $\hat{\rho}_i < 1$, if w_i is large enough and $v_i - w_i$ is small enough, then the employee will not be willing to risk the potential of losing the job offer by negotiating if she has no or low offers from other employers. The remaining difficulty in characterizing the optimal counteroffers is that the reward from negotiating is endogenous to other negotiation decisions. That is, if the employee risks it by negotiating with an employer i, she may optimally decide to not negotiate with an employer j and instead keep it as a backup option.

These complexities are similar to those present in Doval (2018), in which an agent searches by opening "boxes" at a cost to reveal their values, before making a final selection. Importantly, the agent can select a box without opening it. This makes the optimal decision to open a box dependent on the set of available boxes. In our setting, "opening a box" is analogous to making a counteroffer, where the cost is the probabilistic withdrawal of the offer, and choosing not to "open a box" is analogous to settling for the employer's initial wage offer. In both settings, optimal behavior does not follow a (Gittins') index policy, specifically because of

the interdependency of optimal choices on the entirety of the portfolio.

An optimal portfolio of counteroffers always exists despite these complex interdependencies. Moreover, the following proposition finds that for (almost) any collection of parameter values, there is a unique optimal vector of counteroffers.

Proposition 1. *For any given portfolio of offers, there generically (in the space of match-value vectors v) exists a unique optimal vector of counteroffers.*

The proof of Proposition 1 constructs an optimal portfolio by adding employers one at a time, from lowest to highest match value, and at each step considers guaranteed fallback wages. For each possible guaranteed fallback wage, it compares negotiating with the added employer against retaining its initial offer. In both cases, it uses the previously calculated optimal counteroffers to lower-match-value employers, allowing decisions with these employers to vary because retaining the added employer's offer can improve the employee's fallback.

Based on Proposition 1, we henceforth assume uniqueness of the optimal vector of counteroffers, and study three of its properties. First, the employee may optimally negotiate with any subset of employers in her choice set. In other words, an analyst who observes an employee's collection of initial offers can rationalize negotiations with any subset of employers in her portfolio by some collection of match values and beliefs.

Claim 1. *Let $\mathcal{N}$ be a collection of employers, with initial offers $\{w_i\}_{i\in\mathcal{N}}$. Let $\mathcal{N}'$ be an arbitrary, non-empty subset of $\mathcal{N}$. Then there exists a collection of match values and beliefs $\{v_i,\hat{\rho}_i\}_{i\in\mathcal{N}}$ such that the employee optimally negotiates with all employers in the set $\mathcal{N}'$ and no employer in the set $\mathcal{N}\setminus\mathcal{N}'$.*

Reflecting Claim 1, Example 1 in Appendix A.1 demonstrates that it can be optimal for the employee to negotiate with any subset of the two employers making her job offers.

Second, Remark 1 implies the employee will demand a different wage from each employer she makes a counteroffer to, and the following claim additionally implies that an employee will initiate negotiations with (all but one, or all) payoff-relevant employers whose initial offers are equal if she perceives a strictly positive probability that the offer is negotiable. In other words, the employee is not indifferent between the final terms of any two payoff-relevant employers at which negotiation is believed possible.

Claim 2. *The employee has strict preferences over all payoff-relevant, non-rescinded job offers in her portfolio for which negotiation is believed to be possible:* $w_i + o_i^* \neq w_j + o_j^*$ *for all* $i \neq j \in \mathcal{R}(o^*)$ *such that* $\hat{\rho}_i \neq 0$ *and* $\hat{\rho}_j \neq 0$.

Third, optimal counteroffers follow a "top-down" structure. Supposing, without loss of generality, that $w_1 + o_1^* \geqslant w_2 + o_2^* \geqslant \cdots \geqslant w_N + o_N^*$, the employee selects the highest-wage available option, breaking ties arbitrarily (recall by Claim 2 that payoff-relevant employers with strictly positive negotiability beliefs have distinct demanded wages). We show that the employee will make counteroffers to the employers with the highest values, as characterized by the following result.

Proposition 2. *Fix a portfolio of offers such that* $\hat{\rho}_i \in (0,1)$ *for all* $i \in \mathcal{N}$.

1. *Let* $j \in \mathcal{N} \cup \{\varnothing\}$ *be an option with the highest initial offer among those to which the employee optimally does not make a counteroffer, i.e.,* $o_j^* = 0$ *and* $o_i^* > 0$ *for all* $i \in (\mathcal{N} \cup \{\varnothing\}) \setminus \{j\}$ *such that* $w_i > w_j$. *Then the employee optimally negotiates with every employer* ℓ *such that* $v_\ell > v_j$.

2. *For* $i \in \mathcal{R}(o^*)$, *let* $\theta_i := \mathbb{E}\left[\max\left(\{0\} \cup \{\widetilde{w}_j(o^*) : j \in \mathcal{N}, w_j + o_j^* < w_i + o_i^*\}\right)\right]$ *represent the expected final wage of the highest option that is less attractive than* i. *Then the employee makes a counteroffer to* i *if and only if*

$$\theta_i > \frac{w_i - \hat{\rho}_i v_i}{1 - \hat{\rho}_i}.$$

Part 1 describes the "top-down" structure of optimal counteroffers we have previously discussed. Part 2 relates this top-down structure to leverage. When $j \in \mathcal{N}$ and $w_j < v_j$, Part 2 implies that the employee retains the initial offer of the "threshold" employer j in Part 1 because $\hat{\rho}_j v_j + (1 - \hat{\rho}_j)\theta_j < w_j$. The inside option θ_i can be thought of as a notion of leverage at each employer i; the higher θ_i, the less the employee has to lose from negotiating with i. Part 2 therefore states that the higher the inside option, the more leverage the employee has in a negotiation, which increases her desire to negotiate. Indeed, it characterizes how two otherwise "similar" employees who are candidates for the same job at the same employer will make different negotiation decisions—potentially leading to different assignments and furthering inequality—based on different beliefs or fallback options.

Note that while the statements above condition on endogenous events (e.g., the set of employers negotiated with), the proof of Proposition 1 constructs the employee's optimal

equilibrium strategy from model primitives. Therefore, Proposition 2 describes tests an analyst with partial information can take to the data. For example, an analyst may observe (as we do in our observational data) the initial offers a worker receives and counteroffers she makes, but not her match value at employers she does not negotiate with. Part 1 provides information on the relative match value of that employer to others in her portfolio. Part 2 additionally agglomerates several features of the portfolio—beliefs, match values, and initial offers—to bound the "inside option" the employee perceives for each offer.

5.4 Empirical Support & Comparison With Existing Bargaining Models

We evaluate the model against four empirical patterns. Our framework speaks not only to how surplus is divided, but also to whether an employee counters an initial offer and which offers she chooses to negotiate. Our model and evidence point to the employee's beliefs, risk preferences, and portfolio of alternatives as important state variables.

First, bargaining is often employee-initiated and often unsuccessful. Employees frequently countered employers' initial offers, but only 40% of negotiated offers saw improved compensation; the remaining 60% did not. Our model makes the decision to counter explicit and exposes the employee to the possibility that the offer is not negotiable and accompanied by a cost. This combination of active counteroffers and frequent failure is not generated by standard complete-information benchmarks. The starting point for bargaining in many workhorse models of the labor market are Rubinstein (1982) and (Nash, 1950). In the alternating-offer model of Rubinstein (1982), the employer's first offer is accepted in equilibrium. The Nash bargaining solution (Nash, 1950) characterizes the division of surplus but is silent about whether an employee counters an initial offer or how the final surplus split is reached. The presence of proactive counteroffers also contrasts with search and matching models more broadly. Even when the employee has bargaining power, as in Cahuc et al. (2006), employee-initiated negotiations succeed in raising wages whenever they occur in equilibrium, and negotiations only occur with an employer that the employee ends up working for.[62] Our model with incomplete information and employee bargaining power naturally leads to the possibility of failure to reach an agreement.

Second, our experimental treatment increased negotiation attempts by 7.9 pp and compen-

[62] Backus et al. (2020) and Caldwell, Haegele, and Heining (2026) similarly note that the observation of counteroffers in data is evidence against the complete information hypothesis that many models are based on. We additionally observe that many of these attempts fail, which is also not predicted in typical complete information models.

sation gains by 5.7 pp. These effects were concentrated among employees with low prior beliefs about negotiability. The supplemental survey provides direct evidence for the mechanism: the treatment changed respondents' perceived costs and benefits of negotiating, including expected delay, "fit" at the job, and the probability of success (Section 4.2). These findings are difficult to reconcile with complete-information models, which posit that our information treatment contains no payoff-relevant information, and instead support our model's central information-friction hypothesis: information that corrects misperceptions about the efficacy of negotiation can increase both negotiation rates and compensation (Claim 3).[63] Moreover, our model predictions are broadly consistent whether there is only uncertainty about negotiability, only uncertainty about match values, or uncertainty about both. Many standard models of bargaining under incomplete information focus on uncertainty about match values (e.g., Myerson and Satterthwaite, 1983; Chatterjee and Samuelson, 1983; Fudenberg et al., 1985; Fuchs and Skrzypacz, 2010). Our evidence suggests there is also uncertainty about whether an offer is negotiable, a friction that matches how uncertainty is modeled in the reputational bargaining literature (Abreu and Gul, 2000; Atakan and Ekmekci, 2014; Fanning, 2016).[64]

Third, employees with two offers sometimes negotiated the lower offer, sometimes the higher offer, and sometimes both (Panel A of Figure 3). Moreover, the probability of negotiating an offer declines gradually as the gap between that offer and the next-best option grows (Panel B of Figure 3). These patterns reflect our predictions in Claim 1 and Part 2 of Proposition 2, respectively, because negotiation decisions depend on the employee's entire portfolio of options in our model. A strong fallback option insures the employee against negotiation failure, whereas a weak fallback makes risking the top offer less attractive. However, a competing offer is not necessary for negotiation because an employee may still find the expected gain worth the risk (Panel A shows employees with only one offer negotiated 40.5% of the time).[65] By contrast, leverage is often required for negotiation to occur in many bargaining models (see, e.g., Binmore et al. (1989) or Postel-Vinay and Robin (2002a) where low outside options are not credible threats). Because the decision to negotiate depends on risk perceptions and

[63] More formally, Claim 3 in Appendix A.2 shows that an employee with inaccurate prior beliefs benefits when an information treatment corrects those beliefs.

[64] Our evidence suggesting the presence of negotiability uncertainty provides empirical justification for this distinctive modeling choice in the reputational bargaining literature. Moreover, Supplement D.11 presents data consistent with a durable employer reputation. Among the largest employers in our sample, those whose offers were negotiated more often also tended to have somewhat higher negotiation success rates.

[65] We can further restrict to those with the least amount of observable leverage, i.e., employees with only one offer and who were unemployed. A similar share of this subset, 46.5%, negotiated their lone offer (N = 243).

preferences in our model, heterogeneity in the population can generate the gradual decline in the aggregate negotiation rate we observe, as opposed to a common cutoff that would arise if leverage were required for credible bargaining.

Fourth, in our observational data, offer revisions frequently increased the dispersion between options. Consider employees with at least two job options. In the majority of cases, the gap in compensation terms between offers increased when revised, and only 18.9% of respondents had their most competitive final offers within $10,000 of one another (Panel C and Panel D, respectively, of Figure 3). Our model predicts this pattern. Employees may optimally preserve one offer as insurance while negotiating a more attractive offer, so a successful negotiation can widen rather than compress the gap between offers. Indeed, Claim 2 implies that negotiated final wages will typically differ rather than match, while Part 1 of Proposition 2 finds that the highest-match-value employers receive counteroffers. By contrast, models of employer Bertrand competition (e.g., Postel-Vinay and Robin, 2002a,b) predict that competing offers should converge as employers bid for the employee.[66]

Taken together, the evidence supports the three features that distinguish our framework. Employees actively decide whether to counter, those decisions depend on beliefs about negotiability, and negotiations across offers are linked through the insurance provided by the employee's portfolio. Models centered on immediate agreement, complete information, or offer matching do not jointly account for these patterns.

6 Conclusion

Salary negotiations are widespread in some labor markets and have the potential to affect individual job choices, pay equity, and match efficiency. In this paper, we set out to bridge the gap between canonical models of bargaining and the empirical patterns we observe in practice. Using a field experiment among U.S. employees in tech occupations, we find evidence that uncertainty about offer negotiability prevents employees from fully exercising their bargaining power. Broadly, our study provides a framework for understanding why negotiation behavior varies across candidates applying for the same job and highlights the potential benefits of

[66]This prediction of matching offers in Postel-Vinay and Robin (2002a,b) occurs in the limit of full impatience. In general, these models predict that the more productive employer—necessarily the one that ends up matching with the employee—will offer *lower* wages than the less productive employer, because it provides a higher option value in future negotiations. But this finding does not easily comport with our observational finding that employees select the offer with higher compensation 82% of the time. See Section 2.5 for details.

encouraging negotiation.

Our field experiment shows that employees who are provided with facts about negotiation and success rates become more likely to negotiate job offers, and their compensation rises on average. These benefits are concentrated among those who initially underestimated the negotiation rates we shared, and among lower earners. These experimental results, along with supplemental survey responses, suggest employees are uncertain about whether negotiations are permissible, and hence, whether negative repercussions will follow. We embed these findings into a theoretical model of negotiations. The model implies that employees' perceptions of bargaining risk, their tolerance for that risk, and the value of their fallback options together shape bargaining behavior, even though the outcomes they fear, such as a rescinded offer, may rarely be realized. Perceived negotiation risk can therefore exacerbate inequality: holding fixed a set of job offers, those with less visibility into the wage-setting process or who overestimate the risks associated with negotiation are less likely to "push the envelope" and improve their compensation terms. Our experiment and model show that reducing these misperceptions can make employees better off.

Our research also speaks to policy-relevant questions about the role of information in persuasion. *Lean in* campaigns often attempt to persuade individuals to negotiate more (Enwemeka, 2016). Our findings suggest that relatively easy-to-present factual information about the bargaining environment can be a useful way to reduce the perceived risk of negotiation and increase worker surplus. Governments could also reduce bargaining risk directly; for example, they could require employers to make job offers in writing and specify an explicit "no rescission" period in the offer letter. These types of policies could reduce actual and perceived risk of bargaining, encouraging more employees to negotiate. Future research can test the full effects of these policies in practice.

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Table 1: Descriptive Statistics, Attrition, and Randomization Balance

	Full Sample			Follow-Up Sample		
	(1) Encouragement	(2) Control	(3) P-value	(4) Encouragement	(5) Control	(6) P-value
Pre-Treatment Characteristics:						
Age (Years)	31.5	31.3	0.575	30.4	30.4	0.822
	(0.219)	(0.215)		(0.267)	(0.272)	
Share Currently Employed	70.3	71.2	0.629	72.2	71.6	0.807
	(1.317)	(1.311)		(1.723)	(1.833)	
Current/Recent Compensation ($1,000s)	216.4	215.7	0.935	213.6	211.4	0.818
	(6.401)	(6.264)		(7.121)	(6.517)	
Years of Experience	6.9	6.8	0.827	6.3	6.2	0.536
	(0.141)	(0.143)		(0.184)	(0.194)	
Share with Graduate or Professional Degree	52.8	48.7	0.044	50.5	46.9	0.192
	(1.439)	(1.447)		(1.923)	(2.029)	
Share Female	17.7	19.0	0.430	16.0	15.8	0.929
	(1.140)	(1.176)		(1.414)	(1.484)	
Share White	31.5	32.6	0.560	31.2	29.9	0.613
	(1.387)	(1.408)		(1.785)	(1.865)	
Share Asian	47.1	44.6	0.236	52.0	52.8	0.772
	(1.438)	(1.439)		(1.922)	(2.030)	
Share with Baseline Offer	51.6	53.1	0.468	60.1	61.1	0.732
	(1.440)	(1.445)		(1.883)	(1.982)	
Share with Past Negotiation Experience	52.7	52.5	0.928	53.0	50.3	0.335
	(1.439)	(1.446)		(1.920)	(2.033)	
Attrition:						
Responded to Follow-up	66.6	59.7	<0.001			
	(1.360)	(1.420)				
Responded to Follow-up and had 1+ Offers	56.2	50.8	0.008			
	(1.430)	(1.447)				
Observations	1,205	1,194		677	606	

Notes: Average pre-treatment characteristics and follow-up response rates for our main sample, by treatment assignment, with standard errors of the means in parentheses. The left panel (columns (1)–(3)) covers the full randomized sample, and the right panel (columns (4)–(6)) is restricted to respondents who responded to the follow-up survey and reported at least one offer, the sample used in the main analysis. Shares are expressed in percent. *Current/Recent Compensation* is annual base salary, equity, and bonus at the current (or most recent) job. The *Attrition* panel reports follow-up response rates and is defined only for the full sample. P-values test the null hypothesis that the encouragement and control means are equal, and in the *Attrition* panel they test for differential attrition across arms.

Table 2: Treatment Effects on Negotiation Attempts and Compensation Gains

	Main Outcomes					Add. Outcomes	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Intention	Negotiated	Improved	% Improved	$ Improved (000s)	Accepted an Offer	Negotiation Satisfaction
All Respondents	0.063***	0.079***	0.057**	0.999*	1.691	0.005	0.038
	(0.021)	(0.028)	(0.026)	(0.524)	(1.421)	(0.027)	(0.087)
		[0.070**, 0.078**]	[0.060, 0.068**]	[0.851, 1.100**]	[1.368, 2.073]	[0.001, 0.009]	[0.050, 0.059]
Low Prior	0.079***	0.093***	0.074**	1.285**	2.735*	0.024	0.069
	(0.025)	(0.033)	(0.030)	(0.573)	(1.429)	(0.033)	(0.110)
		[0.077*, 0.098***]	[0.073**, 0.087*]	[1.364**, 1.416**]	[3.026*, 3.116**]	[0.004, 0.025]	[0.056, 0.090]
High Prior	0.016	0.038	0.001	0.003	-1.938	-0.042	-0.042
	(0.037)	(0.051)	(0.051)	(1.143)	(3.411)	(0.051)	(0.139)
		[-0.005, 0.080]	[-0.035, 0.051]	[-0.929, -0.043]	[-3.899, -1.688]	[-0.064, 0.022]	[-0.067, 0.047]
P-value: Low vs. High Prior	0.160	0.362	0.217	0.316	0.206	0.270	0.533
Observations	2,321	1,283	1,283	1,283	1,283	1,283	586
Low Prior	1,611	902	902	902	902	902	392
High Prior	710	381	381	381	381	381	194
Control Mean	0.452	0.533	0.300	4.021	9.954	0.591	3.779
Low Prior	0.405	0.511	0.261	3.256	7.191	0.582	3.700
High Prior	0.563	0.590	0.404	6.047	17.278	0.614	3.952

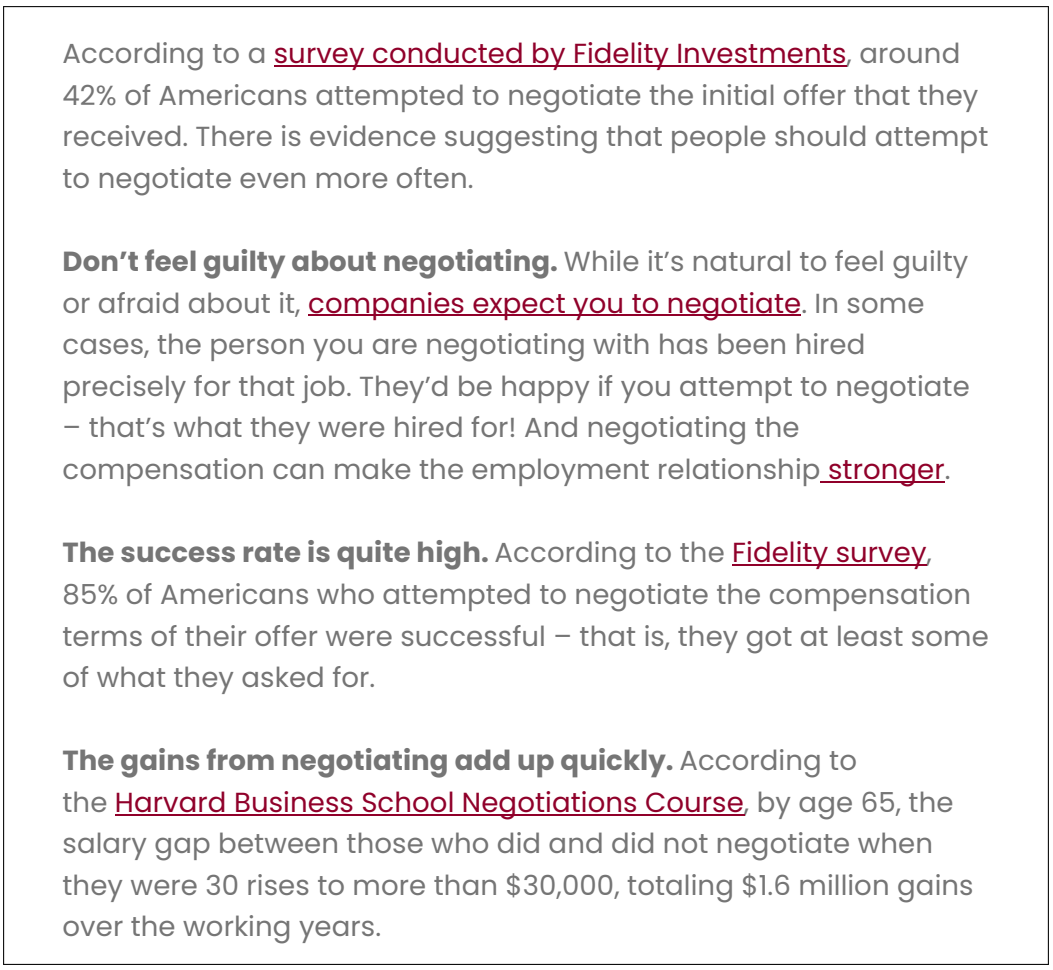
Notes: Column (1) is an indicator for planning to negotiate ("Yes, almost surely"), elicited at baseline and estimated on the full randomized sample who responded to the question. Columns (2)–(7) use the main analysis sample of follow-up respondents with at least one offer. Column (7) is further restricted to respondents who attempted to negotiate at least one offer. Compensation outcomes (columns (4) and (5)) are winsorized at the 99th percentile. Column (6) is an indicator for accepting at least one offer. Column (7) is on a 1–5 scale (1 = "not at all satisfied," 5 = "extremely satisfied"). Low-prior respondents are those with low baseline beliefs about how common and how successful negotiation is. *P-value: Low vs. High Prior* tests equality of the encouragement effect across the two groups. *Control Mean* is the control-group mean of the outcome. Brackets report bounds that account for differential follow-up non-response following Behaghel et al. (2015) (BCGL bounds), for columns (2)–(7). Stars on a bound indicate that it is statistically different from zero. Robust standard errors in parentheses. Significant at *10%, **5%, ***1%.

Table 3: Effects of Ask Size and the Encouragement Treatment (Supplemental Survey)

	(1) Δ Rescind Prob.	(2) Delay (weeks)	(3) Guilt	(4) Risk to Fit	(5) Overall concern	(6) Prob. success
Panel A: 10% vs. 20% Ask						
10% Ask Group Mean (N=276)	5.736***	1.232***	1.083***	1.750***	1.018***	41.322***
	(0.615)	(0.075)	(0.060)	(0.052)	(0.054)	(1.242)
20% Ask Group Mean (N=229)	8.725***	1.323***	1.262***	1.803***	1.258***	35.144***
	(0.933)	(0.074)	(0.068)	(0.061)	(0.067)	(1.303)
Difference	2.989***	0.091	0.179*	0.053	0.240***	-6.178***
	(1.118)	(0.106)	(0.091)	(0.080)	(0.086)	(1.800)
Panel B: Encouragement Treatment						
All Respondents	-1.650	-0.226**	0.164*	-0.243***	0.036	3.884**
	(1.077)	(0.106)	(0.091)	(0.079)	(0.085)	(1.805)
Low Prior (N=365)	-2.018	-0.276**	0.237**	-0.286***	0.136	3.807*
	(1.329)	(0.133)	(0.108)	(0.095)	(0.102)	(2.050)
High Prior (N=140)	-1.086	-0.068	-0.042	-0.148	-0.222	3.392
	(1.946)	(0.169)	(0.172)	(0.146)	(0.150)	(3.567)
P-value: Low vs. High Prior	0.692	0.333	0.169	0.428	0.048	0.920
Control Mean	6.46	1.35	1.01	1.86	0.99	39.72
Low Prior	6.14	1.43	1.02	1.84	1.02	37.50
High Prior	7.37	1.11	1.00	1.92	0.92	46.11
Observations	505	505	505	505	505	505

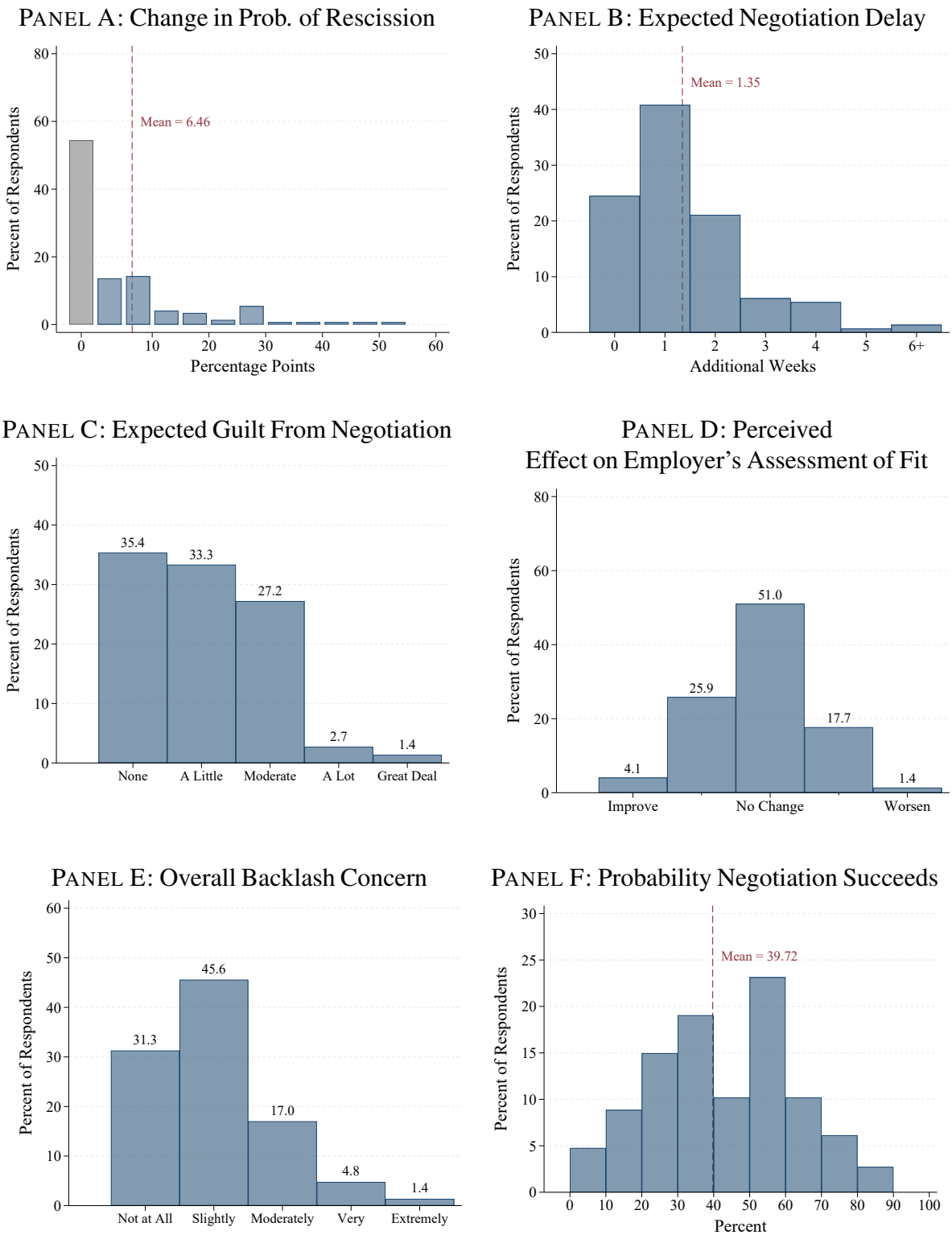
Notes: Supplemental survey analysis sample. Panel A reports means for the 10%- and 20%-ask groups, with t-tests against zero, and their difference, with a Welch t-test. Panel B reports the effect of encouragement, controlling for the 20%-ask indicator. Low-prior respondents are those with low baseline beliefs about how common and how successful negotiation is. *P-value: Low vs. High Prior* is from a model fully interacting prior group with both treatment indicators. *Control Mean* is the mean among respondents assigned the 10% ask and no encouragement. Column (1) is the perceived probability (0–100) that the employer rescinds the offer if the respondent negotiates minus the probability if they do not, in pp, winsorized at the 99th percentile. Respondents who expected negotiating to leave this probability unchanged or lower are coded 0. Column (2) is the expected negotiation delay, in weeks, that negotiating adds to a 2-week start date. The top response, "6+ additional weeks," is coded as 6. Column (3) is expected guilt from negotiating, from 0 (none at all) to 4 (a great deal). Column (4) is the perceived effect on the employer's assessment of the respondent's fit for the role, from 0 (would improve) through 2 (no change) to 4 (would worsen). Column (5) is overall concern about employer backlash, from 0 (not concerned at all) to 4 (extremely concerned). Column (6) is the perceived probability (0–100) that the employer agrees to the new terms. Robust standard errors in parentheses. Significant at *10%, **5%, ***1%.

Figure 1: Encouragement Treatment Message



Notes: This figure provides a screenshot of the message shown to subjects in the encouragement treatment, in both the main experiment and the supplemental survey.

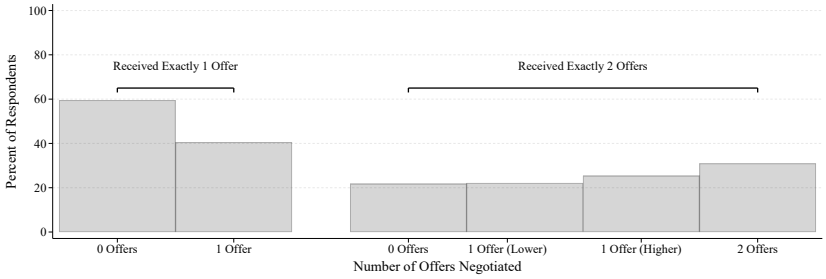
Figure 2: Perceived Costs and Benefits of Negotiating



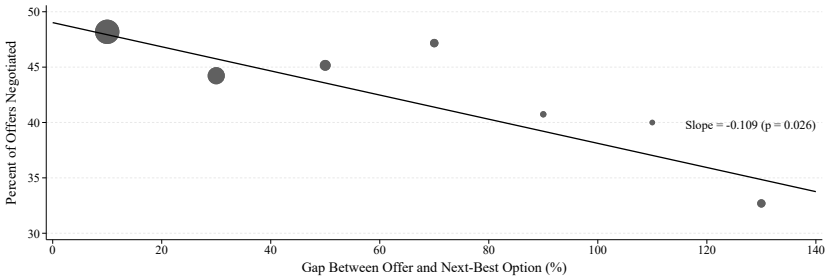
Notes: Answer distributions for the reference group of the supplemental survey: those who did not receive the encouragement treatment and were asked to consider negotiating for a 10% raise (N=147). Panel A is the change in the respondent's stated probability that the offer is rescinded, negotiating versus not, in pp. Respondents who said negotiating would not change or would decrease that probability (2.7% reported a decrease) are coded 0. Values are winsorized at the 99th percentile. The grey bar is the point mass at zero, and other bars are 5-pp bins. For details on the elicitation and construction of this measure, see Supplement E.4. Panel B reports the expected negotiation delay relative to starting in two weeks without negotiating. Panel D shows the perceived effect of negotiating on the employer's assessment of the respondent's fit for the role, relative to not negotiating. Panel E reports respondents' overall backlash concern. Panel F is the perceived probability (0–100) that the employer agrees to the new term.

Figure 3: Empirical Patterns Rationalized by the Bargaining Model

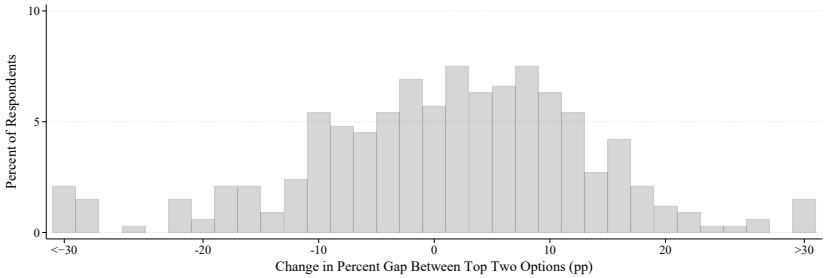
PANEL A: Negotiation With One or Two Offers



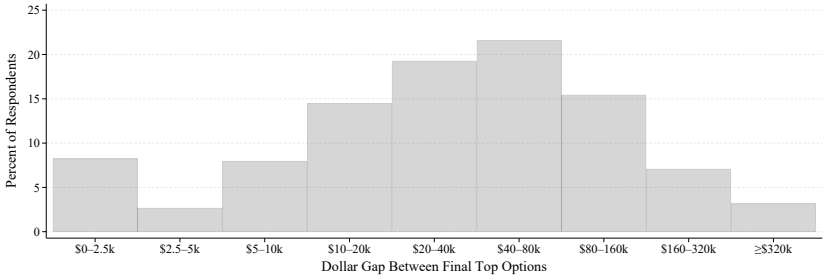
PANEL B: Negotiation by Gap to Next-Best Option



PANEL C: Change in Percent Gap Between Top Two Options



PANEL D: Final Dollar Gap Between Top Two Options



Notes: Panel A shows respondents in our analysis sample with exactly one external job offer (N=896) or exactly two (N=249), by which offers they negotiated. In Panels B–D, a respondent's options are their external offers plus their current employer's compensation, if currently employed. Panel B: The sample is initial offers held by respondents with at least two options, restricted to offers with a lower-valued option of at least $40,000 (N=1,021 offers). The figure plots the share of offers negotiated against the percentage gap between the offer and its next-best option (the highest option below it), in 20-pp bins with marker area proportional to the number of offers, with a fitted linear-probability line (SE clustered by respondent). The gap is winsorized at the 95th percentile. Panel C: The sample is respondents with at least two options for whom a revised offer is among the two highest options at initial or final terms (N=332). The percent gap between the two highest options is their difference as a percentage of their average. The figure plots the change in this gap from initial to final terms, re-ranking the options at each stage, in 2-pp bins. The median change is 2.6 pp. Panel D: The sample is respondents with at least two options at final terms (N=1,042). The figure plots the distribution of the dollar gap between the two highest final options. Bins are approximately equal width on a log scale: edges double from $2,500 to $320,000, with the first bin covering gaps below $2,500 and the last pooling gaps of $320,000 or more.

A Online Appendix: Proofs & Additional Theoretical Results

Proof of Proposition 1:

Proof. Following Remark 1 we restrict to counteroffers $o_i \in \{0, v_i - w_i\}$ for each employer i. A smaller positive counteroffer can be increased to $v_i - w_i$ without changing its acceptance probability, and a larger counteroffer can be replaced by zero without reducing the maximum available wage. Note also that there are finitely many counteroffer vectors such that $o_i \in \{0, v_i - w_i\}$, so a payoff maximizer exists.

Moreover, starting from any such payoff-maximizing counteroffer vector o, a payoff-equivalent vector $\hat{o}$ satisfying our "serious" counteroffers condition exists. To see this, define

$$\hat{o}_i := \begin{cases} o_i, & i \in \mathcal{R}(o), \\ 0, & i \notin \mathcal{R}(o). \end{cases}$$

Compare o and $\hat{o}$ under a given realization of offer negotiability across employers. By construction, the employee almost surely achieves the maximum available wage under o from either the outside option or an employer in $\mathcal{R}(o)$. All such options are unchanged under $\hat{o}$. Thus the maximum available wage weakly increases almost surely. Since o is payoff maximizing, expected payoff is preserved, which implies that the maximum available wage is unchanged almost surely. It remains to show that $\hat{o}$ satisfies the serious-counteroffer restriction. We have already established that every employer in $\mathcal{R}(o)$ remains payoff relevant under $\hat{o}$, i.e., $\mathcal{R}(o) \subseteq \mathcal{R}(\hat{o})$. But note also that every positive counteroffer given $\hat{o}$ is made to an employer in $\mathcal{R}(o)$. Therefore, the vector $\hat{o}$ satisfies the serious-counteroffers condition. Moreover, every positive counteroffer to an employer i in a serious optimum must equal $v_i - w_i$: increasing any smaller positive counteroffer strictly improves payoff on the positive-probability event that i is payoff maximizing.

We now construct an optimal portfolio by induction on the number of employers included. Relabel employers so that $v_1 > \cdots > v_N$. For $k \in \{1,...,N\}$ and $b \geqslant 0$, consider the auxiliary problem in which the employee has offers from exactly employers $k, k+1, ..., N$ and also has a certain outside offer worth b. Employers $1,...,k-1$ are not included in this auxiliary problem. Let $W_k(b)$ denote its highest attainable expected payoff. We will construct both $W_k(b)$ and a coun-

teroffer vector attaining it, for every $b \geqslant 0$. The original problem corresponds to $k=1$ and $b=0$.

As established above, it suffices to consider counteroffers $o_i \in \{0, v_i - w_i\}$. We initially maximize payoff without imposing the serious-counteroffers restriction and apply the serious reduction at the end.

Base Case: No Employers. Initialize at $k=N+1$. The auxiliary problem then contains no employers, so the employee has only the certain outside offer worth b. There are no counteroffers to choose, and her payoff is exactly b. Thus, for every $b \geqslant 0$, set $W_{N+1}(b)=b$. That is, in the zero employer problem, the optimal value is set to equal that of the outside offer b, and it is attained by making no counteroffers.

Induction Step: Add Employer k. Fix $k \in \{1,...,N\}$, and suppose that for every $b \geqslant 0$, we have calculated the optimal expected payoff $W_{k+1}(b)$ from employers $k+1,...,N$ and recorded a vector of counteroffers to those employers that attains it. We now add employer k and solve the problem with employers $k,...,N$.

Fix $b \geqslant 0$. First consider two boundary cases. If $v_k \leqslant b$, then every employer $i \in \{k,...,N\}$ has $w_i \leqslant v_i \leqslant b$. No offer from these employers can improve on the certain outside offer. Hence $W_k(b)=b$, and this payoff is attained by making no counteroffers. If $v_k > b$ and $w_k = v_k$, making no counteroffer to k guarantees v_k, and no other employer in this auxiliary problem can offer more. Hence $W_k(b)=v_k$, again attained by making no counteroffers. The remaining case is that $v_k > b$ and $w_k < v_k$. There are two possible choices regarding employer k:

Do not make a counteroffer to employer k. Setting $o_k=0$ retains w_k with certainty. Given outside offer b, this gives a guaranteed wage of $\max\{b,w_k\}$. The employee's problem of choosing counteroffers to employers $k+1,...,N$ is therefore exactly the auxiliary problem with that guaranteed wage. By the induction hypothesis, the highest attainable expected payoff from this choice is

$$W_{k+1}(\max\{b,w_k\}), \tag{A.1}$$

which is attained by setting $o_k=0$ and using the counteroffers already recorded for employers $k+1,...,N$ at guaranteed wage $\max\{b,w_k\}$.[67]

[67] Recall that the induction hypothesis is that we have calculated $W_{k+1}(b)$ for every value of b. Letting $b' = \max\{b,w_k\}$ therefore implies that the optimal value $W_{k+1}(b')$ has been calculated, and counteroffers o_ℓ

Make a counteroffer to employer k**.** That is, $o_k = v_k - w_k$. If accepted, this gives payoff v_k, regardless of the offers available from employers $k+1,...,N$, because their match values and the certain outside offer b are all strictly below v_k. If rejected, the employee chooses between the certain outside offer b and the available offers from employers $k+1,...,N$. By the induction hypothesis, their highest attainable expected payoff together with the certain outside offer is $W_{k+1}(b)$ prior to adding employer k. Thus the highest expected payoff from making a counteroffer to employer k is

$$\hat{\rho}_k v_k + (1-\hat{\rho}_k) W_{k+1}(b), \tag{A.2}$$

which is attained by setting $o_k = v_k - w_k$ and using the counteroffers already recorded for employers $k+1,...,N$ at guaranteed wage b. The same formula applies when $\hat{\rho}_k = 1$, since the rejection term then has zero weight.

These two choices exhaust the possibilities for employer k. Each of the expressions in equations (A.1) and (A.2) is both an upper bound on the payoff attainable under its respective choice and attained by the counteroffer vector just described. Consequently,

$$W_k(b) = \max\{W_{k+1}(\max\{b, w_k\}), \hat{\rho}_k v_k + (1-\hat{\rho}_k) W_{k+1}(b)\}.$$

Record the counteroffer vector associated with $W_k(b)$, breaking equality in favor of no counteroffer to employer k. This constructs the optimal payoff and a vector attaining it for the problem with employers $k,...,N$. Since b was arbitrary, the induction step holds for every $b \geqslant 0$. Note that electing not to make a counteroffer to k may change the optimal counteroffers to employers $k+1,...,N$, because those counteroffers are then chosen using the recorded solution for guaranteed wage $\max\{b, w_k\}$ rather than b.

Conclusion. Starting from the zero-employer base case and applying the induction step for $k = N, N-1, ..., 1$ constructs an optimal counteroffer vector for every auxiliary problem. At $k = 1$ and $b = 0$, the auxiliary problem is the original portfolio problem. The recorded counteroffer vector o' attaining $W_1(0)$ is therefore payoff maximizing. Following the argument at the beginning of the proof, it is then possible to impose the serious counteroffer restriction on o'—i.e., by reducing o'_i to zero for any $i \notin \mathcal{R}(o')$ to generate an optimal, serious counteroffer vector o^*.

to employers $\ell \in \{k+1,...,N\}$ are known. Retain such counteroffers, and add $o_k = 0$ to attain $W_{k+1}(b')$ in the problem expanded with employer k.

We note that although the induction is stated as requiring us to solve $W_k(b)$ for all $b \geqslant 0$, the calculation of $W_1(0)$ requires only the guaranteed wages in the finite set $\{w_i : i \in \mathcal{N} \cup \{\varnothing\}, w_i \geqslant 0\}$. Indeed, the only possible update to a guaranteed wage is $b \mapsto \max\{b, w_k\}$, which preserves membership in this set. Thus one can initialize $W_{N+1}(b) = b$ for each value in this set and perform the induction step for each such b at $k = N, N-1, \ldots, 1$. There are therefore at most $N(N+1)$ comparisons between making and not making a counteroffer.

It remains to prove generic uniqueness. Fix $(w_i, \hat{\rho}_i)_{i \in \mathcal{N}}$ and endow the feasible match-value space $\mathcal{V} := \{v \in \mathbb{R}^N : v_i \geqslant w_i \text{ for every } i, v_i \neq v_j \text{ for distinct } i,j\}$ with its relative Euclidean topology. Define $\mathcal{V}_0 := \{v \in \mathcal{V} : v_i \neq 0 \text{ for every } i, v_i \neq w_j \text{ for every } i,j\}$. For $v \in \mathcal{V}_0$ and a set of employers $D \subseteq \mathcal{N}$, let $o^D(v)$ denote the counteroffer vector that negotiates with exactly the employers in D:

$$o_i^D(v) := \begin{cases} v_i - w_i, & i \in D, \\ 0, & i \notin D. \end{cases}$$

Because $v \in \mathcal{V}_0$ implies $v_i > w_i$ for every i, distinct sets D give distinct counteroffer vectors. Let $U_D(v)$ denote the employee's expected payoff from $o^D(v)$ at match-value profile v, and define $\mathcal{D}(v) := \{D \subseteq \mathcal{N} : D \subseteq \mathcal{R}(o^D(v))\}$. Thus $D \in \mathcal{D}(v)$ precisely when every employer receiving a positive counteroffer under $o^D(v)$ is payoff relevant. The collection $\mathcal{D}(v)$ is finite and contains the empty set.

Define

$$\mathcal{G} := \left\{ v \in \mathcal{V}_0 : \begin{array}{l} \text{there exists } D \in \mathcal{D}(v) \text{ such that} \\ U_D(v) > U_{D'}(v) \text{ for every } D' \in \mathcal{D}(v) \setminus \{D\} \end{array} \right\}.$$

We show three properties about $\mathcal{G}$ (in relation to $\mathcal{V}$). First, that these strict payoff comparisons imply uniqueness of the serious optimum. Second, that it is open. Third, that it is dense.

Uniqueness on $\mathcal{G}$. Fix $v \in \mathcal{G}$ and let D^* satisfy the strict payoff comparisons in the definition of $\mathcal{G}$. We have already established that a serious payoff maximizer exists and that every positive counteroffer in a serious optimum equals $v_i - w_i$. Consequently, every serious optimum is represented by some $D \in \mathcal{D}(v)$, and maximizing $U_D(v)$ over this finite collection attains the global optimal payoff. The defining inequalities for $\mathcal{G}$ imply that D^* is the unique maximizing set in this collection. Therefore $o^{D^*}(v)$ is the unique serious optimal counteroffer vector.

Openness. We first establish that, in a sufficiently small neighborhood of each $v \in \mathcal{V}_0$, the collection of serious negotiation sets is unchanged and the expected payoff from every fixed negotiation set varies continuously with match values. Let

$$r(v) := \frac{1}{4}\min\left(\{|v_i| : i \in \mathcal{N}\} \cup \{|v_i - w_j| : i,j \in \mathcal{N}\} \cup \{|v_i - v_j| : i,j \in \mathcal{N},\, i \neq j\}\right) > 0.$$

If $\|v' - v\|_\infty < r(v)$, all ordinal comparisons between match values, initial wages, and zero are unchanged. In particular, $v' \in \mathcal{V}_0$. Therefore, for any D the set of payoff-maximizing available employers is the same in every realization of negotiability at both v and v'. Hence $\mathcal{D}(v') = \mathcal{D}(v)$ whenever $\|v' - v\|_\infty < r(v)$. Moreover, under the same negotiability realizations, each available wage changes by at most $\|v' - v\|_\infty$. The maximum available wage therefore changes by at most this amount. Taking expectations gives, for every $D \subseteq \mathcal{N}$, $|U_D(v') - U_D(v)| \leqslant \|v' - v\|_\infty$.

Now fix $v \in \mathcal{G}$, and let D^* be its unique serious optimal negotiation set. If D^* is the unique element of $\mathcal{D}(v)$ then the fact that $\mathcal{D}(v') = \mathcal{D}(v)$ when v' is in some neighborhood of v immediately implies that $v' \in \mathcal{G}$. Therefore, consider that D^* is not the unique element of $\mathcal{D}(v)$. Define the payoff surplus of D^* by $g(v) := \min_{D \in \mathcal{D}(v) \setminus \{D^*\}} \left[U_{D^*}(v) - U_D(v)\right] > 0$, where the strict inequality follows from the uniqueness of D^* for $v \in \mathcal{G}$. For every v' satisfying $\|v' - v\|_\infty < \min\{r(v), g(v)/4\}$, the collection of serious negotiation sets is unchanged, and for every $D \in \mathcal{D}(v) \setminus \{D^*\}$, $U_{D^*}(v') - U_D(v') \geqslant g(v) - 2\|v' - v\|_\infty > g(v)/2 > 0$. Thus, the same D^* is uniquely optimal for v', which establishes that $\mathcal{G}$ is open in $\mathcal{V}$.

Denseness. We will show that $\mathcal{G}$ is dense in $\mathcal{V}$ by proving that its complement is nowhere dense. There are exactly two reasons a profile can lie outside $\mathcal{G}$: a match value equals zero or an initial wage, or two distinct serious negotiation sets attain the same maximal payoff. To account for these possibilities, define $\mathcal{B}_0 := \mathcal{V} \setminus \mathcal{V}_0$. For each pair of distinct sets $D, D' \subseteq \mathcal{N}$, define

$$\mathcal{B}_{D,D'} := \left\{ v \in \mathcal{V}_0 : \begin{array}{l} D, D' \in \mathcal{D}(v), \\ U_D(v) = U_{D'}(v), \\ U_D(v) \geqslant U_E(v) \text{ for every } E \in \mathcal{D}(v) \end{array} \right\}.$$

Because $\mathcal{D}(v)$ is nonempty and finite, failure of a unique maximizing negotiation set means that at least two distinct sets attain its maximal payoff. Therefore,

$$\mathcal{V}\setminus\mathcal{G}=\mathcal{B}_0\cup\bigcup_{\substack{D,D'\subseteq\mathcal{N}\\ D\neq D'}}\mathcal{B}_{D,D'}.$$

There are finitely many sets in this union. We show that each is nowhere dense.

First, $\mathcal{B}_0$ is closed in $\mathcal{V}$, since it is a finite union of sets defined by equalities $v_i = 0$ or $v_i = w_j$. It also has empty interior. To see this, fix any $v \in \mathcal{V}$ and define, for $t > 0$, $v' := (v_1+t, v_2+t, ..., v_N+t)$. $v' \in \mathcal{V}$ because the inequalities $v'_i \geqslant w_i$ are preserved, as are all differences $v'_i - v'_j = v_i - v_j \neq 0$. Only finitely many values of t yield $v_i + t = 0$ or $v_i + t = w_j$ for any i,j. Thus $v' \in \mathcal{V}_0$ for every sufficiently small $t > 0$. Consequently, $\mathcal{B}_0$ is nowhere dense in $\mathcal{V}$.

Next, fix distinct D, D'. On $\mathcal{V}_0$, $\mathcal{D}(v)$ is constant in some neighborhood of v and all payoffs $U_E(v)$ are continuous, as established in the openness argument. Therefore $\mathcal{B}_{D,D'}$ is closed relative to $\mathcal{V}_0$ as it is locally defined by one payoff equality and finitely many weak payoff inequalities. We show that $\mathcal{B}_{D,D'}$ has empty interior in $\mathcal{V}_0$. Fix $v \in \mathcal{B}_{D,D'}$. Without loss of generality, suppose $D \setminus D'$ is non-empty; as we have assumed that $D \neq D'$, if this is not the case, then $D' \setminus D$ is non-empty, and the subsequent argument can easily be adjusted to accommodate. Choose $i \in D \setminus D'$. Because D is serious, $i \in \mathcal{R}(o^D(v))$. The employee demands wage v_i under $o^D(v)$, which by construction differs from zero, every initial wage, and every other match value. Thus, whenever employer i's final terms are payoff maximizing, it is the unique highest-paying available option. Let $p_i^D > 0$ denote the probability of this event. For $0 < \varepsilon < r(v)$, define v' by increasing only v_i to $v_i + \varepsilon$ and leaving all other match values unchanged. Under negotiation set D, the counteroffer to employer i increases by ε, so its demanded wage remains its full match value. All wage comparisons and acceptance probabilities are unchanged. The employee's realized payoff increases by exactly ε with probability p_i^D and remains unchanged with probability $1 - p_i^D$. That is, $U_D(v') = U_D(v) + p_i^D \varepsilon$. Under D', employer i is not negotiated with, so its available wage remains w_i and $U_{D'}(v') = U_{D'}(v)$. Since $U_D(v) = U_{D'}(v)$, it follows that $U_D(v') - U_{D'}(v') = p_i^D \varepsilon > 0$. Thus $v' \notin \mathcal{B}_{D,D'}$. Because v' can be defined for any sufficiently small ε, $\mathcal{B}_{D,D'}$ has empty interior in $\mathcal{V}_0$.

Fix any nonempty open set $\mathcal{U} \subseteq \mathcal{V}$. Because $\mathcal{V}_0$ is open and dense in $\mathcal{V}$, $\mathcal{U} \cap \mathcal{V}_0$ is nonempty and open. Choose $v \in \mathcal{U} \cap \mathcal{V}_0$. If $v \notin \mathcal{B}_{D,D'}$, set $v' = v$. Otherwise, apply the single-coordinate perturbation to construct v', choosing $\varepsilon > 0$ sufficiently small that $v' \in \mathcal{U} \cap \mathcal{V}_0$. We have established that $v' \notin \mathcal{B}_{D,D'}$. Thus, in either case, $v' \in (\mathcal{U} \cap \mathcal{V}_0) \setminus \mathcal{B}_{D,D'}$. Because $\mathcal{B}_{D,D'}$ is closed relative to $\mathcal{V}_0$, there is an open neighborhood $\mathcal{U}'$ of v' contained in $\mathcal{U} \cap \mathcal{V}_0$ such that $\mathcal{U}' \cap \mathcal{B}_{D,D'} = \varnothing$.

Thus every nonempty open set in $\mathcal{V}$ contains a nonempty open neighborhood in which D and D' are not both serious optimal negotiation sets. This establishes that $\mathcal{B}_{D,D'}$ is nowhere dense in $\mathcal{V}$.

Since there are finitely many pairs D, D', the complement $\mathcal{V} \setminus \mathcal{G}$ is a finite union of nowhere-dense sets and is therefore nowhere dense. Hence $\mathcal{G}$ is dense in $\mathcal{V}$. Together with openness and uniqueness on $\mathcal{G}$, this proves generic uniqueness.

□

Proof of Claim 1

Proof. For any $\mathcal{N}$ and $\mathcal{N}'$, we construct vectors v and $\hat{\rho}$ that generate the desired negotiation behavior. Let $W = \max(\{0\} \cup \{w_i : i \in \mathcal{N}\})$. For every $i \in \mathcal{N}'$, choose distinct match values $v_i > 2W+1$ and set $\hat{\rho}_i = 1/2$. For every $i \notin \mathcal{N}'$, set $\hat{\rho}_i = 0$ and choose distinct match values for each employer i satisfying $v_i > w_i$. Only employers in $\mathcal{N}'$ can receive serious positive counteroffers.

Therefore, it remains to show that the employee negotiates with all $i \in \mathcal{N}'$. Suppose for contradiction that at a serious optimal counteroffer vector o some $i \in \mathcal{N}'$ is not negotiated with. Let $J \subset \mathcal{N}'$ be the set of employers receiving a strictly positive counteroffer under o. Define counteroffer vector o' by adding a counteroffer for employer i, i.e.,

$$o'_j := \begin{cases} v_i - w_i, & j = i, \\ o_j, & j \neq i. \end{cases}$$

Whenever at least one employer in J accepts its counteroffer, the employee has an offer paying strictly more than employer i's initial wage w_i. Moving from o to o' changes only the counteroffer to employer i, so all offers from employers in J remain unchanged. Thus, putting employer i's initial offer at risk cannot lower the employee's payoff in these realizations. Therefore, condition on the realizations that the offers of all employers in J are non-negotiable, which occurs with probability $1/2^{|J|} > 0$. Under o, the final payoff is at most W in such realization, whereas the conditional expected payoff under o' is at least $v_i/2 > W$. Thus o' yields a strictly higher expected payoff than does o. Contradiction with the assumption that $i \in \mathcal{N}'$ is not negotiated with. Thus, we have established, as desired, that every employer in $\mathcal{N}'$ must therefore be negotiated with in an optimum, and no employer outside $\mathcal{N}'$ is negotiated with. □

Proof of Claim 2

Proof. Consider the unique serious optimal counteroffer vector o^*. Suppose, toward a contradiction, that distinct employers $i,j \in \mathcal{R}(o^*)$ with $\hat{\rho}_i, \hat{\rho}_j > 0$ satisfy $w_i + o_i^* = w_j + o_j^*$. There are three exhaustive cases, each leading to a contradiction:

- *Both employers receive counteroffers.* We have established that every positive counteroffer in a serious optimum equals the employer's match value minus its initial wage. Thus $w_i + o_i^* = v_i$ and $w_j + o_j^* = v_j$, so the proposed tie contradicts $v_i \neq v_j$.
- *Neither employer receives a counteroffer.* Then $w_i = w_j$. At least one employer, say i, has $v_i > w_i$, for otherwise $v_i = v_j$. Making the counteroffer $v_i - w_i$ to employer i while retaining employer j's offer as a sure fallback cannot lower payoff. Payoff relevance and $\hat{\rho}_i > 0$ imply a strictly positive probability that the original maximum wage is $w_i = w_j$ and employer i accepts the counteroffer. Thus the change strictly improves expected payoff, contradicting optimality.
- *One employer receives a counteroffer.* Without loss of generality, $o_i^* > 0$ and $o_j^* = 0$. This implies $v_i = w_j$. Consider an alternative counteroffer vector o that differs only in that $o_i = 0$. By construction o preserves the maximum available wage in every realization of negotiability compared to o^*, implying that o is also payoff maximizing. Removing any potential unserious counteroffers from o (as in the proof of Proposition 1) results in a distinct serious counteroffer vector $\hat{o}$ that achieves the same payoff as o^*. Contradiction with o^* being the unique serious optimal counteroffer vector.

□

Proof of Proposition 2

Proof of Part 1: Suppose, toward a contradiction, that $v_\ell > v_j$ but $o_\ell^* = 0$. By the definition of j, $w_\ell \leqslant w_j \leqslant v_j < v_\ell$. Consider an alternative o' which coincides with o^* except that it counteroffers $v_\ell - w_\ell$ to employer ℓ. o' cannot lower the realized payoff relative to o^* since w_j remains available for certain. Because $\hat{\rho}_i \in (0,1)$ for all i, all negotiated offers except employer ℓ's are rescinded and ℓ accepts the counteroffer with positive probability under o'. In such cases, the realized payoff increases from w_j to v_ℓ, contradicting optimality of o^*. □

Proof of Part 2: Fix $i \in \mathcal{R}(o^*)$. If $w_i = v_i$, then $o_i^* = 0$, and payoff relevance implies $w_i \geqslant 0$. Thus $\theta_i \leqslant w_i = (w_i - \hat{\rho}_i v_i)/(1-\hat{\rho}_i)$, establishing the claim. Henceforth, take $w_i < v_i$.

By Claim 2, payoff-relevant employers have distinct demanded wages. For each i, write $x_i^* := w_i + o_i^*$. The event that all offers with demanded wages strictly above x_i^* are unavailable has strictly positive probability. Conditional on this event, the expected highest available wage among employers with demanded wages strictly below x_i^* (including the outside option) is θ_i.

Any alternative attaining the optimal payoff can be reduced to a serious optimum as in the proof of Proposition 1. Suppose first that $o_i^* > 0$, so $x_i^* = v_i$. Conditional on the event above, the expected payoff is $\hat{\rho}_i v_i + (1-\hat{\rho}_i)\theta_i$. Keeping all other counteroffers fixed and instead choosing not to negotiate with i yields at least w_i at this event and leaves payoff unchanged whenever a higher-wage offer is available. Therefore, uniqueness of the optimal serious counteroffer vector implies $\hat{\rho}_i v_i + (1-\hat{\rho}_i)\theta_i > w_i$.

Conversely, suppose $o_i^* = 0$. Consider keeping all other counteroffers fixed and instead counteroffering $v_i - w_i$ to employer i. When an offer above w_i is available, this alternative cannot lower payoff. When all such offers are unavailable, the alternative yields conditional expected payoff $\hat{\rho}_i v_i + (1-\hat{\rho}_i)\theta_i$ instead of w_i. At the positive-probability event that employer i accepts the counteroffer and all offers above w_i are unavailable, employer i is the unique highest-paying option (and therefore, i remains payoff relevant after removing counteroffers to payoff irrelevant employers). As before, the unique optimality of o^* implies $\hat{\rho}_i v_i + (1-\hat{\rho}_i)\theta_i < w_i$.

Rearranging the strict inequalities terminating the previous two paragraphs establishes the result. □

A.1 Employee May Optimally Negotiate Any Subset of Offers

As discussed in Section 5.3, negotiation with any subset of employers in an employee's portfolio can be rationalized. The following example demonstrates this, and in doing so shows that finding an optimal portfolio of counteroffers may involve non-trivial considerations, as it is insufficient to "locally" optimize—negotiating with an employer has consequences on the marginal risk courted by the rest of the portfolio.

Example 1. *Let* $\mathcal{N} = \{1,2\}$*, with* $v_1 > v_2$*,* $0 < w_i < v_i$*, and* $\hat{\rho}_i \in (0,1)$*. By Remark 1, negotiation entails demanding the full surplus from that employer, and there are thus four relevant counteroffer vectors.* U_{00} *represents the expected payoff from negotiating neither of the offers,* U_{10}

from negotiating only the first, U_{01} *from negotiating only the second, and* U_{11} *from negotiating both. Then*

$$U_{00} = \max\{w_1, w_2\}, \qquad U_{10} = \hat{\rho}_1 v_1 + (1-\hat{\rho}_1) w_2,$$
$$U_{01} = \hat{\rho}_2 \max\{w_1, v_2\} + (1-\hat{\rho}_2) w_1, \qquad U_{11} = \hat{\rho}_1 v_1 + (1-\hat{\rho}_1)\hat{\rho}_2 v_2.$$

U_{11} *is largest when both negotiability beliefs are sufficiently close to one.* U_{10} *is largest when* $\hat{\rho}_2 v_2 < w_2$ *and* $\hat{\rho}_1$ *is sufficiently close to one. If* $w_1 > v_2$, U_{00} *is largest when* $\hat{\rho}_1$ *is sufficiently small, as no offer from employer 2 can improve on retaining* w_1.

If $v_2 > w_1$, *fix* $\hat{\rho}_2 < 1$ *sufficiently close to one. Then* U_{01} *is strictly largest when* $\hat{\rho}_1 > 0$ *is sufficiently small given* $\hat{\rho}_2$. *To see this, as* $\hat{\rho}_1 \to 0$, $U_{10} \to w_2$ *and* $U_{11} \to \hat{\rho}_2 v_2$, *whereas* $U_{01} = \hat{\rho}_2 v_2 + (1-\hat{\rho}_2) w_1$ *strictly exceeds both limits and* U_{00}. *The employee therefore retains employer 1's initial offer as insurance while negotiating with employer 2.*

As demonstrated in the last comparison, the value of retaining an offer as insurance can depend on negotiation decisions with other employers. To make this explicit, let $(v_1, v_2) = (10, 9)$, $(w_1, w_2) = (5, 4)$, *and* $(\hat{\rho}_1, \hat{\rho}_2) = (2/5, 3/5)$. *Then* $(U_{00}, U_{10}, U_{01}, U_{11}) = (5, 6.4, 7.4, 7.24)$. *Any "rule" that first implies negotiation with employer 2 because* $\hat{\rho}_2 v_2 = 5.4 > w_2$, *and then implies negotiation with employer 1 because* $\hat{\rho}_1 v_1 + (1-\hat{\rho}_1)\hat{\rho}_2 v_2 = 7.24 > w_1$, *would incorrectly prescribe negotiating with both. Retaining employer 1's offer does not preclude receiving* v_2 *from employer 2: it instead guarantees* w_1 *if that negotiation fails. Negotiating only with employer 2 is strictly optimal.*

A.2 Model Rationalizes Experimental Finding on Initial Misperceptions

Experimental evidence shows that employees who initially have the most misspecified beliefs about the viability of bargaining (i.e., those who are pessimistic about negotiation efficacy) experience the largest gains from receiving accurate information. This short section rationalizes that employees with inaccurate beliefs benefit from bargaining under corrected beliefs.

Consider a portfolio $\{w_i, v_i, \hat{\rho}_i\}_{i \in N}$ under which the employee engages in bargaining. Suppose that her beliefs are incorrect, i.e., the vector governing employer negotiability is $\rho \neq \hat{\rho}$. Clearly, searching under incorrect beliefs $\hat{\rho}$ leads to an optimal vector of counteroffers that may not be optimal. By contrast, an information treatment that corrects her beliefs to ρ will lead to the optimal expected value maximizing vector of counteroffers. The following simple claim states this finding in the context of our experiment in which employees differ in their beliefs.

Claim 3. *Suppose an information treatment (for example, one similar to that in our experiment) induces correct beliefs about the viability of negotiation in all employees. Then an employee with misperception in her prior belief will weakly benefit relative to an otherwise-identical employee with accurate prior beliefs.*

Proof. Write $U_\rho(o)$ for payoff evaluated under the true probabilities under counteroffer vector o. Since $o^*(\rho)$ maximizes this objective, $U_\rho(o^*(\rho)) \geqslant U_\rho(o^*(\hat{\rho}))$. □

A.3 Extension: Match-Value Uncertainty

This appendix considers an extension of our base model in which the employee has uncertainty about match values with employers. Our qualitative predictions continue to hold under this extension. Proofs of these results are housed in Supplement B.

Fix a finite set of employers $\mathcal{N} := \{1,...,N\}$. Each employer $i \in \mathcal{N}$ makes a (known) initial wage offer $w_i \in \mathbb{R}$ to an employee. Each employer has a match value v_i. The employee does not know the match value v_i at each employer. Instead, she knows that $v_i \in [w_i, \bar{v}_i]$, where $\bar{v}_i > w_i$, and that the *incremental surplus* $s_i := v_i - w_i \in [0, \bar{s}_i]$, $\bar{s}_i := \bar{v}_i - w_i$, is distributed according to a cumulative distribution function F_i on $[0, \bar{s}_i]$. We refer to this form of uncertainty as *(match) value* uncertainty.

As in the baseline model in Section 5 of the paper, the employee potentially has uncertainty about whether each offer is negotiable. She has beliefs $\hat{\rho} = (\hat{\rho}_1,...,\hat{\rho}_N), \hat{\rho}_i \in [0,1]$, where $\hat{\rho}_i$ is the employee's belief that employer i's offer is negotiable. If the employee makes a counteroffer to a non-negotiable employer, the offer is withdrawn. If the employee makes a counteroffer to a negotiable employer, it is accepted if and only if it does not exceed the match value. All random variables, namely each offer's negotiability and surplus draw, are mutually independent.

Therefore, a portfolio of offers from employers in $\mathcal{N}$ can be represented $\{w_i, \bar{v}_i, \hat{\rho}_i, F_i\}_{i \in \mathcal{N}}$. Again, we endow the worker with an outside option $\varnothing$ where $w_\varnothing = \bar{v}_\varnothing = o^*_\varnothing = 0$. We make a number of assumptions pertaining to the portfolio of offers.

1. $\hat{\rho}_i > 0$ for all employers i. Otherwise, the employer j with the largest initial offer among employers who make non-negotiable offers with probability 1 (i.e. $j \in \underset{i \in \mathcal{N}}{\operatorname{argmax}} \{w_i | \hat{\rho}_i = 0\}$) provides a de facto outside option $b = \max\{w_j, 0\}$ for the employee and our results hold for portfolio $\{w_i - b, \bar{v}_i - b, \hat{\rho}_i, F_i\}_{i \in \mathcal{N} \setminus \{j\}}$.

2. For each employer i, $\bar{v}_i > 0$. Otherwise, a job at i would be less attractive than the outside option with probability 1 and will play no role in our upcoming analysis,

3. $\bar{v}_i \neq \bar{v}_j$ for $i \neq j \in \mathcal{N}$, which is an analogue of the genericity condition that $v_i \neq v_j$ for $i \neq j \in \mathcal{N}$ from the model presented in the body of the paper, and

4. Finally, we assume, as is standard, that the distributions are smooth and Myerson regular:

   Assumption 1. *For each $i \in \mathcal{N}$, F_i is twice continuously differentiable with associated density f_i such that $f_i(o_i) > 0$ for all $o_i \in [0, \bar{s}_i]$. We further assume that F_i is* (strictly) Myerson regular;[68] *the virtual value function*

   $$\phi_i(o_i) := o_i - \frac{1 - F_i(o_i)}{f_i(o_i)}$$

   satisfies $\phi_i'(o_i) > 0$ for every $o_i \in [0, \bar{s}_i]$.

   For each $i \in \mathcal{N}$, let $\mathcal{F}_i$ denote the set of distributions on $[0, \bar{s}_i]$ satisfying Assumption 1. Let $\mathcal{F} := \prod_{i=1}^{N} \mathcal{F}_i$.

The employee chooses an *additional wage demand* (i.e., a counteroffer increment) $o_i \in \{0\} \cup \mathbb{R}_{++}$ for each employer i.

- If $o_i = 0$, the employee does *not* initiate negotiation with i, and the offer remains available at wage w_i.

- If $o_i > 0$, the employee initiates negotiation with i and requests wage $w_i + o_i$. The offer remains available at wage $w_i + o_i$ with probability $\hat{\rho}_i \cdot \Pr(s_i \geqslant o_i) = \hat{\rho}_i(1 - F_i(o_i))$, and is withdrawn otherwise.

Given a portfolio $\{w_i, \bar{v}_i, \hat{\rho}_i, F_i\}_{i \in \mathcal{N}}$ and any counteroffer vector o, the wage available from each employer i is a random variable given by

[68] Note that "strict Myerson regularity" is often described as $\phi(\cdot)$ being strictly increasing. Since F_i is twice continuously differentiable and $f_i > 0$ on the compact interval $[0, \bar{s}_i]$, the virtual value ϕ_i is continuously differentiable. Our condition $\phi_i'(o_i) > 0$ for every $o_i \in [0, \bar{s}_i]$ is stronger than strict monotonicity of ϕ: by compactness, it implies $\inf_{o_i \in [0, \bar{s}_i]} \phi_i'(o_i) > 0$, while strict monotonicity of ϕ does not rule out points where $\phi_i'(o_i) = 0$ or is arbitrarily close to zero. This additional technical structure is used only in the proof of Theorem 1, where it ensures that Myerson regularity is preserved under sufficiently small smooth perturbations of F_i in the $\mathcal{C}^2$ norm.

$$\widetilde{w}_i(o)=\begin{cases} w_i, & \text{if } o_i=0,\\ w_i+o_i, & \text{with probability } \hat{\rho}_i(1-F_i(o_i)) \text{ if } o_i>0,\\ -\infty, & \text{otherwise if } o_i>0. \end{cases}$$

Let $\widetilde{w}_\varnothing(o)=0$ denote the outside option for all o. The employee is hired by an employer $i^*\in \underset{i\in\mathcal{N}\cup\{\varnothing\}}{\operatorname{argmax}}\, \tilde{w}_i$, i.e. an employer with the highest available offer (ties are broken arbitrarily, as doing so does not affect payoffs).

We consider the same objective (to maximize the expected payoff) and restriction (to consider only "serious" counteroffers) as in the base model presented in the body of the paper. Specifically, for a given portfolio of wage offers $\{w_i,\bar{v}_i,\hat{\rho}_i,F_i\}_{i\in\mathcal{N}}$, the employee seeks a vector of counteroffers, one to each employer, that satisfies:

1. Optimal counteroffers $\{o_i^*\}_{i\in\mathcal{N}}$ maximize the employee's expected payoff, and

2. Only "serious" counteroffers are made; if the employee has a zero probability of matching with an employer $i\in\mathcal{N}$ under any selection rule according to criterion 1, then $o_i^*=0$. Formally, employer i is *payoff relevant* under o if it belongs to the set of payoff-maximizing available options with strictly positive probability.

$$\mathcal{R}_F(o):=\left\{i\in\mathcal{N}:\Pr\left(i\in\underset{j\in\mathcal{N}\cup\{\varnothing\}}{\operatorname{argmax}}\widetilde{w}_j(o)\right)>0\right\}$$

denotes the set of payoff-relevant employers under o. Only serious counteroffers at an optimal o^* require $o_i^*=0$ if $i\in\mathcal{N}\setminus\mathcal{R}_F(o^*)$.

A.3.1 Optimal Counteroffer Against Fixed Alternative

Here, we consider a building block in which the employee faces an offer from a single employer i and has a fixed outside option $\theta_i\in\mathbb{R}$ if she fails to reach an agreement with i. We assume $\bar{v}_i>\theta_i\geqslant 0$, otherwise the employee will never accept a position at employer i even under the maximum possible wage. Thus, the "serious" counteroffers restriction implies that no negotiation will occur.

The employee makes two related choices. First, conditional on negotiating ($o_i>0$), what is the optimal counteroffer? Second, is the expected value generated from the optimal counteroffer sufficient to overcome the risk of offer withdrawal?

We begin by addressing the first question: what is the employee's optimal, strictly positive counteroffer? For $o_i + w_i \geqslant \theta_i$ with $o_i > 0$, the employee's expected payoff is

$$\begin{aligned} U_i(o_i;\theta_i) &= [1-\hat{\rho}_i(1-F_i(o_i))]\theta_i + \hat{\rho}_i(1-F_i(o_i))(w_i+o_i) \\ &= \theta_i + \hat{\rho}_i(1-F_i(o_i))(w_i+o_i-\theta_i). \end{aligned} \tag{A.3}$$

Note that $U_i(o_i;\theta_i)$ is defined only for $o_i > 0$, because negotiation necessarily involves making a strictly positive counteroffer. It will be useful in the sequel to extend its domain by defining $U_i(0;\theta_i) := \theta_i + \hat{\rho}_i(w_i - \theta_i)$, i.e., to allow the employee to make a zero counteroffer while still experiencing a $1-\hat{\rho}_i$ risk of offer rescission. This representation is useful to make compact the set of possible counteroffers, but will not represent the true payoff at $o_i = 0$. Our presented results do not rely on an employee negotiating by making a zero counteroffer.

The following lemma characterizes the optimal *interior* counteroffer $o_i^{int} \in (0, \bar{s}_i)$:

Lemma 1. *Consider maximizing* $U_i(\cdot;\theta_i)$ *over* $o_i \in (0,\bar{s}_i]$. *If* $w_i - \theta_i \geqslant \frac{1}{f_i(0)}$, *then there is no interior maximizer. If* $w_i - \theta_i < \frac{1}{f_i(0)}$, *the unique interior maximizer* $o_i^{int}(\theta_i)$ *satisfies:*

$$w_i + o_i^{int}(\theta_i) - \theta_i = \frac{1-F_i(o_i^{int}(\theta_i))}{f_i(o_i^{int}(\theta_i))}. \tag{A.4}$$

We now address the second question: is the expected value generated from the optimal counteroffer sufficient to overcome the non-negotiability risk $1-\hat{\rho}_i$? Define the *stage value* as the largest expected payoff the employee can obtain from employer i when her continuation value if not employed by i is θ_i:

$$V_i(\theta_i) := \max\left\{ w_i, \max_{o_i \in [0,\bar{s}_i]} U_i(o_i;\theta_i) \right\}. \tag{A.5}$$

The first term, w_i is the payoff from being hired by employer i at the initial wage. The second term is the best expected payoff from initiating negotiation with employer i. Under the maintained assumptions that $\theta_i < \bar{v}_i$ and $\hat{\rho}_i > 0$, $\max_{o_i \in [0,\bar{s}_i]} U_i(o_i;\theta_i) > \theta_i$; moreover, $U_i(0;\theta_i)$ cannot strictly improve on not negotiating because $U_i(0;\theta_i) \leqslant \max\{w_i,\theta_i\}$. Thus the employee initiates negotiation if and only if $V_i(\theta_i) > w_i$, where throughout, we break indifference in favor of not negotiating. In words, this condition is satisfied if and only if the risk of offer withdrawal is sufficiently low to justify negotiating. Note two "sanity checks." First, if $\theta_i \geqslant w_i$, then the employee will always be better off making some counteroffer for any $\hat{\rho}_i > 0$. To see this, note that because $\theta_i < \bar{v}_i = w_i + \bar{s}_i$, the interval $(\theta_i - w_i, \bar{s}_i)$

is well defined. For any $o_i \in (\theta_i - w_i, \bar{s}_i)$, we have $w_i + o_i > \theta_i$ and $1 - F_i(o_i) > 0$, so $U_i(o_i;\theta_i) = \theta_i + \hat{\rho}_i(1-F_i(o_i))(w_i+o_i-\theta_i) > \theta_i \geqslant w_i$. Thus, $V_i(\theta_i) > w_i$. Second, if $w_i - \theta_i \geqslant \frac{1}{f_i(0)}$, then Lemma 1 implies that $U_i(\cdot;\theta_i)$ has no interior maximizer. Indeed, $U_i(\cdot;\theta_i)$ is strictly decreasing on $(0,\bar{s}_i)$, so $\max_{o_i \in [0,\bar{s}_i]} U_i(o_i;\theta_i) = U_i(0;\theta_i) = \theta_i + \hat{\rho}_i(w_i - \theta_i) \leqslant w_i$. Hence, under the convention that indifference is broken in favor of not negotiating, negotiation is not optimal.

To formalize the optimal decision rule, for any $\Delta \in \mathbb{R}$, define

$$r_i(\Delta) := \max_{o_i \in [0,\bar{s}_i]} (1-F_i(o_i))(\Delta + o_i).$$

Since $\max_{o_i \in [0,\bar{s}_i]} U_i(o_i;\theta_i) = \theta_i + \hat{\rho}_i r_i(w_i - \theta_i)$, the employee negotiates with employer i if and only if $\hat{\rho}_i r_i(w_i - \theta_i) > w_i - \theta_i$. Combining this negotiation condition with Lemma 1 yields the following takeaway:

Remark 2. *The optimal counteroffer (where we break indifference in favor of not negotiating) is given by:*

$$o_i^*(\theta_i) = \begin{cases} o_i^{\text{int}}(\theta_i), & \text{if } \hat{\rho}_i r_i(w_i - \theta_i) > w_i - \theta_i, \\ 0, & \text{otherwise.} \end{cases}$$

The following finds that leverage matters in this single-employer problem; a higher outside option implies a more aggressive counteroffer.

Lemma 2. o_i^* *is weakly increasing in* θ_i *on* $[0,\bar{v}_i)$.

A.3.2 Portfolio Problem: Existence and (Generic) Uniqueness

We now return to the original problem involving a finite set of employers $\mathcal{N} = \{1,...,N\}$ associated with a portfolio of offers $\{w_i,\bar{v}_i,\hat{\rho}_i,F_i\}_{i\in\mathcal{N}}$. Let $o = \{o_i\}_{i=1,...,N}$ represent the vector of counteroffers. Define the *demanded wages* $x_i := w_i + o_i$ and the *availability probabilities*

$$a_i(o_i) := \begin{cases} 1, & o_i = 0, \\ \hat{\rho}_i(1-F_i(o_i)), & o_i > 0. \end{cases}$$

The employee's payoff is given by the highest available wage, including the outside option 0.

Let π be a permutation of $\{1,...,N\}$ such that $x_{\pi(1)} \geqslant x_{\pi(2)} \geqslant \cdots \geqslant x_{\pi(N)}$. The employee selects the highest available nonnegative wage according to this ordering, or takes the outside

option if none is available. Ties ($x_i = x_j$ for $i \neq j$) do not affect the expected payoff because the employee receives the same wage whichever tied offer has a lower index in the permuted order, i.e., is selected with higher precedence. The expected payoff from o is therefore

$$U(o) = \sum_{k=1}^{N} \max\{x_{\pi(k)}, 0\} a_{\pi(k)}(o_{\pi(k)}) \prod_{m<k} \left(1 - a_{\pi(m)}(o_{\pi(m)})\right). \tag{A.6}$$

We use the above construction to argue that an optimal vector of counteroffers exists for any portfolio.

Proposition 3. *Fix a finite set of employers $\mathcal{N} = \{1,...,N\}$ associated with a portfolio of offers $\{w_i, \bar{v}_i, \hat{\rho}_i, F_i\}_{i \in \mathcal{N}}$. The employee has an optimal counteroffer vector*

$$o^* \in \prod_{i \in \mathcal{N}} [0, \bar{s}_i].$$

A.3.3 A Constructive Characterization Conditional on an Ordering

We now turn to a recursive construction of a candidate optimal counteroffer vector, which we later use to argue the uniqueness of the optimal vector of counteroffers generically in the space of portfolios. We construct these candidates by first fixing the set of employers receiving counteroffers and deriving wage offers recursively according to a proposed ordering.

Fix a portfolio $\{w_i, \bar{v}_i, \hat{\rho}_i, F_i\}_{i \in \mathcal{N}}$. Let $D \subseteq \mathcal{N}$ and let π be a proposed ordering of the employers in D from highest to lowest demanded wage. Write $m = |D|$ and $b_D := \max\left(\{0\} \cup \{w_j : j \notin D\}\right)$. Set $o_j^{D,\pi} = 0$ for $j \notin D$ and initialize the fallback option given employers not negotiated with, $\theta_m = b_D$. For $k \in \{m, m-1, ..., 1\}$, let $i = \pi(k)$ and define

$$o_i^{D,\pi}(F) := \underset{z \in [0, \bar{s}_i]}{\operatorname{argmax}} \{\theta_k + \hat{\rho}_i (1 - F_i(z))(w_i + z - \theta_k)\}. \tag{A.7}$$

Including employer $\pi(k)$ updates the continuation value to

$$\theta_{k-1} := \theta_k + \hat{\rho}_i (1 - F_i(o_i^{D,\pi}))(w_i + o_i^{D,\pi} - \theta_k). \tag{A.8}$$

For every pair (D, π) and every F, the maximizer in equation (A.7) is unique. If $\theta_k < \bar{v}_i$, Lemma 1 gives either a unique interior maximizer or finds the unique maximizer to be zero. If $\theta_k \geqslant \bar{v}_i$, the unique maximizer is $\bar{s}_i$: every smaller demand yields an objective function value below θ_k. Therefore, $\theta_{k-1} \geqslant \theta_k$, because choosing $z = \bar{s}_i$ yields $\theta_{k-1} = \theta_k$. Note that

the constructed wage demands $\{w_i + o_i^{D,\pi}\}_{i \in D}$ need not correspond to order π, however, our following definition restricts to candidate counteroffer vectors that correspond to the proposed ordering π.

Definition 1. *The candidate counteroffer vector for* (D,π) *and* F *defined by equations (A.7) and (A.8) is* regular *if* $0 < o_i^{D,\pi}(F) < \bar{s}_i$ *for every* $i \in D$ *and* $w_{\pi(1)} + o_{\pi(1)}^{D,\pi} > \cdots > w_{\pi(m)} + o_{\pi(m)}^{D,\pi} > b_D$. *For* $D = \varnothing$*, let the zero counteroffer vector be regular.*

As established later in our genericity argument, we consider only regular candidate counteroffer vectors. For regular counteroffer vectors, $V^{D,\pi}(F) := \theta_0 = U(o^{D,\pi}(F))$. Moreover, every employer in D is payoff relevant: its wage exceeds b_D and all higher-ranked negotiated offers have positive rejection probabilities. Thus every retained candidate vector satisfies the serious counteroffer restriction, where $o_i^{D,\pi} > 0$ implies

$$w_i + o_i^{D,\pi} - \theta_k = \frac{1 - F_i(o_i^{D,\pi})}{f_i(o_i^{D,\pi})}, \qquad i = \pi(k). \tag{A.9}$$

Let Γ be the finite set of all pairs (D,π), including the empty ordering for $D = \varnothing$, and let $\Gamma_F \subseteq \Gamma$ contain the pairs (D,π) whose constructed counteroffer vectors are regular at F. The following result establishes that comparing these candidates solves the original portfolio problem.

Lemma 3. *For every* $F \in \mathcal{F}$*, every serious optimal counteroffer vector is represented by a pair in* Γ_F*. Consequently,* Γ_F *is nonempty, and maximizing* $V^{D,\pi}(F)$ *over* $(D,\pi) \in \Gamma_F$ *yields a serious optimum.*

Proposition 3 establishes existence of a serious optimum, and Lemma 3 represents every such optimum within a finite set of alternatives based on candidate orderings. We now show that, generically in the distribution profile, exactly one regular candidate corresponding to some (D,π) has the highest expected payoff.

To do so, we must formalize genericity in the space of distributions. Recall that for each employer i, $\mathcal{F}_i$ represents the set of distributions on $[0, \bar{s}_i]$ satisfying Assumption 1. Let $\mathcal{F} := \prod_{i=1}^{N} \mathcal{F}_i$. Endow each $\mathcal{F}_i$ with the weak-$*$ topology (which is equivalent to weak convergence of the associated probability measures on the compact interval $[0,\bar{s}_i]$), and endow $\mathcal{F}$ with the product topology. A property $\mathcal{P}$ is said to hold *generically* if it holds on an *open and dense* subset of $\mathcal{F}$ (in the product topology).

Theorem 1. *Fix* $(w_i,\bar{v}_i,\hat{\rho}_i)_{i \in \mathcal{N}}$*. There exists an open and dense subset* $\mathcal{G}^{ext} \subseteq \mathcal{F}$ *such that for every* $F = (F_1,\ldots,F_N) \in \mathcal{G}^{ext}$*:*

1. *the employee's serious optimal counteroffer portfolio* $o^\star(F)$ *is unique; and*

2. *the induced demanded wages are tie-free among payoff-relevant employers:* $w_i + o_i^\star(F) \neq w_j + o_j^\star(F)$ *for all distinct* $i,j \in \mathcal{R}_F(o^\star(F))$.

We now state the analogue of Proposition 2 from the paper for the present value-uncertainty extension. Throughout this section, assume we are in the "generic case," i.e., $F \in \mathcal{G}^{\text{ext}}$ from Theorem 1. Let $o^\star$ be the unique optimal serious counteroffer vector and define demanded wages $x_i^\star := w_i + o_i^\star$.

As in the baseline model, for each payoff-relevant employer i define its *inside option* by

$$\theta_i := \mathbb{E}\left[\max\left(\{0\} \cup \{\widetilde{w}_j(o^*) : j \in \mathcal{N},\, x_j^* < x_i^*\}\right)\right].$$

For o^* derived from a pair $(D,\pi) \in \Gamma_F$, and an employer i with $o_i^* > 0$ and $i = \pi(k)$, the inside option θ_i calculated here equals the value θ_k constructed in equation (A.8).

Proposition 4. *Let* $F \in \mathcal{G}^{\text{ext}}$ *and let* $o^\star$ *be the unique serious optimal counteroffer vector at* F.

1. *Let* $j \in \mathcal{N} \cup \{\varnothing\}$ *be an option with the highest initial wage among those to which the employee optimally does* not *make a counteroffer:* $o_j^\star = 0$ *and* $o_i^\star > 0$ *for all* $i \neq j$ *with* $w_i > w_j$. *Then the employee makes a counteroffer to every employer* ℓ *such that* $\bar{v}_\ell > \bar{v}_j$.

2. *The employee makes a counteroffer to employer* $i \in \mathcal{R}_F(o^*(F))$ *if and only if*

$$V_i(\theta_i) > w_i, \tag{A.10}$$

   where V_i *is the stage-value function defined in equation (A.5).*

3. *If the employee makes a counteroffer to employer* $i \in \mathcal{R}_F(o^*(F))$ *(i.e.* $o_i^\star > 0$*) then* $o_i^\star$ *satisfies*

$$(w_i + o_i^\star) - \theta_i = \frac{1 - F_i(o_i^\star)}{f_i(o_i^\star)}. \tag{A.11}$$